Numerical-relativity particle-in-cell method at low particle number: An independent reconstruction by language-model agents

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Provisional abstract of a draft in progress; results marked [pending] were not final when it was written. Collisionless matter near black holes obeys the Einstein-Vlasov system, where particles follow geodesics of their curvature. Wu and Most introduced a numerical-relativity particle-in-cell method with millions of phase-space packets called macroparticles. We ask whether their text enables an independent rebuild and what changes at ten thousand macroparticles. Language-model agents under human direction built the code from text alone in AthenaK, with separate models for code, checks, and writing. Predictions written before each control run tested every diagnosis. Each step evolves spacetime like a field solver, advances macroparticles by a pusher, and spreads sources onto cells by cloud-in-cell deposition. The implicit midpoint pusher matches Kerr orbits, while the spacetime Boris pusher needs an unstated local-frame tetrad gauge: symmetric analytically, Cholesky on meshes. Tracers near fixed and evolving black holes conserve invariants, and deposition conserves sources across refinement. Geometric slice-mass checks of apparent horizons, outermost surfaces of non-expanding light rays, exposed an unfilled ghost-cell defect in AthenaK's apparent-horizon finder. Interpolant derivatives agree within the accuracy of the mass measure. Surface mass rescaling removes dust-collapse lattice excess at ten thousand macroparticles. A filter compensation pass cancels cloud widening that expands stable clusters. Level-capped deposition coarsens clouds yet worsens constraints and apparent-horizon mass drift near black holes, so finest-level deposition stays. Dust collapses to a black hole of matching apparent-horizon mass while lapse excision removes interior particles frozen by time dilation. Monoenergetic clusters, equal-energy stellar spheres, stay bound with a growing core oscillation. The cause of that growth and coarsened-cloud drift is not determined. Past the binding-energy maximum, unperturbed unstable clusters expand toward migration [pending], perturbed ones collapse [pending], and unbound ones disperse [pending]. Prolate clouds collapse [pending], and a relativistic cluster with two collapsing overdensities at speed-consistent radius merges [pending]. These controls establish a reproducible algorithm: ten thousand macroparticles capture black hole formation and cluster stability.

I. INTRODUCTION

Stars in dense relativistic clusters and collisionless dark matter around black holes move freely through collective gravitational fields that vary strongly in space and time [1–5]. Two-body gravitational encounters act over timescales far longer than a dynamical crossing time, much as binary collisions remain negligible in a collisionless plasma [2, 6]. Multi-stream trajectories cross in configuration space and build anisotropic stresses, whereas single-fluid hydrodynamics assigns one bulk velocity to each spatial point [3, 7]. Strong-field evolution therefore requires the Einstein-Vlasov system, a phase-space kinetic description coupled to the full Einstein equations where matter follows geodesics of the curvature its stress-energy generates [3, 8, 9]. Early numerical studies with spherical codes mapped equilibrium sequences, radial stability, and black hole formation in relativistic clusters [1, 10–12] alongside critical gravitational collapse [13–15]. Subsequent axisymmetric and three-dimensional simulations extended kinetic gravity to rotating clusters, nonspherical collapse, and cosmic structure [7, 16–20], where asymmetric instabilities and mergers demand evolution in three spatial dimensions.

The numerical-relativity particle-in-cell (NRPIC) method of Ref. [21] brings algorithms from relativistic plasma simulation [6, 22–25] to three-dimensional Einstein-Vlasov evolution. Macroparticles—finite-mass packets sampling six-dimensional phase space—advance along timelike geodesics through a time-reversible trajectory pusher in the reconstructed metric [21, 26–29]. Conservative cloud-in-cell deposition assigns each macroparticle's energy, momentum, and stress to nearby mesh cells across refinement boundaries before grid filtering smooths discrete sampling noise [6, 21, 22, 30]. A constraint-damped solver in moving-puncture gauge, the standard way to evolve Einstein's equations through black hole formation on a fixed grid, advances the spacetime on an adaptive mesh [21, 31–34]. Reference [21] demonstrated this coupled cycle on gravitational collapse and relativistic cluster benchmarks using millions of macroparticles on large machines.

Two questions decide how broadly this method can be adopted. One question asks whether an independent builder without access to the original code can reconstruct the NRPIC method quantitatively from its published description alone. A numerical method is usable by others only when its written equations and algorithmic rules suffice for an independent implementation to recover its benchmarks. The second question asks which results survive at ten thousand macroparticles, where a run takes minutes to a few hours on one graphics card, and what must change for that. A small computational entry cost decides who can use the method for routine physical exploration, rapid parameter surveys, and independent verification.

Language-model agents under human direction built an independent implementation from the published text alone inside the public AthenaK code [35, 36], with portable parallel execution through Kokkos [37]. Every algorithmic stage and coupled test was compared with exact analytical solutions or with the published numbers of Ref. [21]. Candidate explanations of unexpected results were tested against predictions recorded before control runs designed so that they could fail, as described in Sec. VI.

The published description suffices to rebuild the method and reproduce the single-particle errors of Ref. [21] once one unstated choice is identified (Sec. IV). That choice is the tetrad gauge, the local rotation freedom of the orthonormal frame in the spacetime Boris pusher, where distinct frame constructions shift the discrete error (Secs. III A and IV). Self-gravitating evolutions at ten thousand macroparticles require two algorithmic changes (Sec. V). Surface masses on a cubic dust lattice must be rescaled because discrete site counts inside a sphere exceed its continuum volume at low particle number (Sec. V A). Multi-pass source filters require a sharpening compensation pass [6] because plain smoothing widens clouds like gravitational softening and makes stable clusters bounce outward (Secs. III C and V C). Coarser particle clouds—a tempting change on sparse refined meshes—raise constraint violations and drive black hole mass drift, so clouds belong on the finest local level (Sec. V). A geometric mass measure of the slice also exposed a ghost-zone defect in the host code's finder for apparent horizons, the outermost closed surfaces of zero outgoing light expansion (Secs. III F and V A). With these elements in place, ten thousand macroparticles reproduce in kind dust collapse [38], the sequence of relativistic clusters [1, 12], and the migration of an unstable cluster toward the stable branch [21, 39] (Secs. V A and V C). Documented differences from Ref. [21] appear in the collapse times of unstable clusters and in the collapse of one perturbed near-critical model (Sec. V C).

Section II formulates collisionless matter in a dynamical spacetime and its macroparticle phase-space representation. Section III presents the particle-mesh method: geodesic pushers, metric reconstruction, stress-energy deposition and filtering, spacetime evolution, adaptive mesh refinement, and numerical methods. Section IV tests the geodesic pushers, phase mixing, source conservation, and interior particle removal by lapse excision around fixed and evolving black holes. Section V studies self-gravitating matter at ten thousand macroparticles across spherical dust collapse and the sequence of relativistic clusters. Section VI describes how the code was assembled and verified, before the final section states our conclusions and the appendix records the scenario parameters.

II. COLLISIONLESS MATTER IN A DYNAMICAL SPACETIME

Stars in dense clusters, particles in dark-matter spikes, and debris in colliding stellar streams move freely through smooth collective gravitational fields [1–3]. Two-body gravitational encounters act over timescales far longer than a dynamical crossing time, so each constituent follows a geodesic of the spacetime geometry generated by the whole ensemble. Throughout this paper we set Newton's gravitational constant G and the speed of light c to unity and adopt metric signature $(-,+,+,+)$. Greek letters μ,ν,ρ,σ label spacetime indices, Latin letters i,j,k,l label spatial coordinate indices, and parenthesized indices $(m),(n),(k)$ label components in a tetrad, a local frame of four mutually perpendicular unit vectors.

Collisionless matter in a curved spacetime with metric $g_{\mu\nu}$ and inverse $g^{\mu\nu}$ is described in phase space by a rest-mass-weighted distribution function $f(x^\mu,u^\nu)$ [3, 8, 9]. Here x^μ are spacetime coordinates and u^μ is the four-velocity, equal to the four-momentum per unit rest mass on the future mass shell $g_{\mu\nu}u^\mu u^\nu=-1$. Phase-space conservation along free-fall worldlines coupled to Einstein's field equations forms the Einstein-Vlasov system,

$$u^\mu\partial_\mu f-\Gamma^\mu_{\nu\rho}u^\nu u^\rho\frac{\partial f}{\partial u^\mu}=0,\qquad G_{\mu\nu}=8\pi T_{\mu\nu},\tag{1}$$

where $\Gamma^\mu_{\nu\rho}$ are the Christoffel symbols of $g_{\mu\nu}$. The Einstein tensor $G_{\mu\nu}$ is sourced by the stress-energy tensor $T_{\mu\nu}$, given by the second four-velocity moment of f over the future mass shell.

Time evolution splits the spacetime into spacelike hypersurfaces of constant coordinate time t with spatial coordinates x^i, induced metric γ_{ij}, inverse γ^{ij}, and determinant $\gamma=\det(\gamma_{ij})$ [40]. Local measurements on each slice are referred to normal observers, Eulerian observers whose worldlines cross the slice orthogonally with future-directed unit four-velocity n^μ and covector $n_\mu=(-\alpha,0,0,0)$. The lapse α links adjacent slices by converting a coordinate interval $\mathrm{d}t$ into proper time $\alpha\,\mathrm{d}t$ along normal observers. The shift β^i gives the coordinate velocity at which spatial grid points drift relative to those normal observers. In these variables the spacetime line element reads

$$\mathrm{d}s^2=-\alpha^2\,\mathrm{d}t^2+\gamma_{ij}(\mathrm{d}x^i+\beta^i\,\mathrm{d}t)(\mathrm{d}x^j+\beta^j\,\mathrm{d}t),\tag{2}$$

with $n^\mu=(\alpha^{-1},-\alpha^{-1}\beta^i)$. The bending of each slice inside the ambient spacetime is measured by the symmetric extrinsic curvature K_{ij}, defined through

$$\partial_t\gamma_{ij}=-2\alpha K_{ij}+D_i\beta_j+D_j\beta_i,\tag{3}$$

where D_i is the spatial covariant derivative compatible with γ_{ij} and $\beta_i=\gamma_{ij}\beta^j$. Normal observers project $T^{\mu\nu}$ into ten matter fields—the energy density E, covariant momentum density S_i, and symmetric spatial stress

S_{ij}—according to

$$E = n_\mu n_\nu T^{\mu\nu}, \qquad S_i = -g_{i\mu} n_\nu T^{\mu\nu}, \\ S_{ij} = g_{i\mu} g_{j\nu} T^{\mu\nu}. \tag{4}$$

Direct grid discretization of f spans six phase-space dimensions and becomes prohibitive in three spatial dimensions without symmetry, whereas the Vlasov equation conserves phase-space weight along timelike geodesics. A particle-in-cell method [6, 21, 22] exploits these characteristics by sampling f with N finite-weight phase-space packets called macroparticles, indexed by $p = 1, \ldots, N$. With $u_\mu = g_{\mu\nu} u^\nu$, each macroparticle carries fixed rest mass m_p, spatial coordinates $x^i_p(t)$, and covariant spatial four-velocity components $u_{i,p}(t) = g_{i\mu} u^\mu_p$, giving six phase-space variables per particle. On the future mass shell $g^{\mu\nu} u_\mu u_\nu = -1$, the Lorentz factor seen by normal observers is

$$W = \alpha u^0 = \sqrt{1 + \gamma^{ij} u_i u_j}, \tag{5}$$

which determines the single-particle Hamiltonian $-u_0 = \alpha W - \beta^i u_i$. Hamilton's canonical equations for (x^i, u_i) yield the characteristic equations,

$$\frac{\mathrm{d}x^i}{\mathrm{d}t} = \frac{\alpha \gamma^{ij} u_j}{W} - \beta^i, \\ \frac{\mathrm{d}u_i}{\mathrm{d}t} = -W \partial_i \alpha + u_j \partial_i \beta^j - \frac{\alpha}{2W} u_j u_k \partial_i \gamma^{jk}. \tag{6}$$

At spatial coordinate $\boldsymbol{x} = (x^1, x^2, x^3)$, the ensemble stress-energy projected onto normal observers forms the particle sources,

$$E(t, \boldsymbol{x}) = \sum_{p=1}^{N} \frac{m_p W_p}{\sqrt{\gamma}} \, \delta^3(\boldsymbol{x} - \boldsymbol{x}_p), \\ S_i(t, \boldsymbol{x}) = \sum_{p=1}^{N} \frac{m_p u_{i,p}}{\sqrt{\gamma}} \, \delta^3(\boldsymbol{x} - \boldsymbol{x}_p), \\ S_{ij}(t, \boldsymbol{x}) = \sum_{p=1}^{N} \frac{m_p u_{i,p} u_{j,p}}{W_p \sqrt{\gamma}} \, \delta^3(\boldsymbol{x} - \boldsymbol{x}_p). \tag{7}$$

Here $\delta^3(\boldsymbol{x} - \boldsymbol{x}_p)$ is the three-dimensional coordinate Dirac distribution and W_p is evaluated at $\boldsymbol{x}_p(t)$. Geodesic transport of $(x^i_p, u_{i,p})$ under the characteristic equations, coupled to smoothed mesh projections of the particle sources, solves the Einstein-Vlasov system.

III. THE PARTICLE-MESH METHOD

Coupled Einstein-Vlasov evolution advances smooth curvature fields on an Eulerian mesh alongside discrete trajectories over a shared global time step Δt, set by the Courant limit of the finest cell spacing Δx. The numerical-relativity particle-in-cell (NRPIC) method [21] links both representations in a staggered cycle, analogous to electromagnetic particle-in-cell codes where Maxwell's equations take the place of the spacetime solver [6, 22]. Finite sampling with a mean count N_{ppc} of macroparticles per cell introduces sampling noise, discrete fluctuations of relative amplitude $N_{\mathrm{ppc}}^{-1/2}$. Within this cycle, the geodesic pushers preserve orbital time-reversal symmetry. Unfiltered deposition conserves particle weight across refinement boundaries. Subsequent grid filtering smooths the deposited sources at modest N_{ppc}.

One cycle from t^n to $t^{n+1} = t^n + \Delta t$ runs through five stages on a closed loop, with mesh regridding on a side branch (Fig. 1). First, the spacetime stage saves α, β^i, and γ_{ij} at t^n across active and ghost cells. Four fourth-order Runge–Kutta stages then advance the curvature variables to t^{n+1} at fixed sources (E, S_i, S_{ij}) from t^n. Second, cubic Hermite reconstruction supplies the metric fields and their spatial gradients averaged between t^n and t^{n+1}. Third, a geodesic pusher—an update advancing each macroparticle's position and momentum across Δt—pushes every particle from t^n to t^{n+1} in that averaged metric. Fourth, interior excision removes particles deep inside black holes, discards out-of-domain or nonfinite states, and reassigns surviving particles to their enclosing blocks. Fifth, stress-energy deposition assigns each macroparticle's mass-energy, momentum, and stress to nearby cells at t^{n+1}, filters (E, S_i, S_{ij}), and evaluates the constraints. Diagnostics then run. On the regrid branch, the adaptive mesh refines or coarsens where curvature gradients cross thresholds. After any regrid, metric arrays, block ownership, and sources are rebuilt on the new leaf cells while $(m_p, x^i_p, u_{i,p})$ stay unchanged.

A. Geodesic pushers

Collisionless equilibria and close black hole passages require trajectory updates free of secular drift in orbital invariants. The implicit midpoint pusher and the spacetime Boris pusher [21] store x^i_p and $u_{i,p}$ at integer time levels, reach second-order accuracy in Δt, and preserve time-reversal symmetry through midpoint evaluation. Each queries the geometry solely through a point-evaluation function returning α, β^i, γ_{ij}, and their first spatial derivatives, so identical code runs on analytic metrics, uniform lattices, and adaptive meshes. Fixed-point iteration updates the six endpoint variables $(x^{i,n+1}_p, u^{n+1}_{i,p})$ until the largest component change falls below a tolerance.

The implicit midpoint pusher evaluates the characteristic equations at the phase-space midpoint (with the particle index p suppressed),

$$\bar{x}^i = \frac{x^{i,n} + x^{i,n+1}}{2}, \qquad \bar{u}_i = \frac{u^n_i + u^{n+1}_i}{2}. \tag{8}$$

Spatial reconstructions at t^n and t^{n+1} are evaluated at $\bar{\boldsymbol{x}}$ and averaged to obtain $\bar{\alpha}$, $\bar{\beta}^i$, $\bar{\gamma}_{ij}$, and their spatial derivatives. Matrix inversion of $\bar{\gamma}_{ij}$ yields $\bar{\gamma}^{ij}$, $\bar{u}^i = \bar{\gamma}^{ij} \bar{u}_j$,

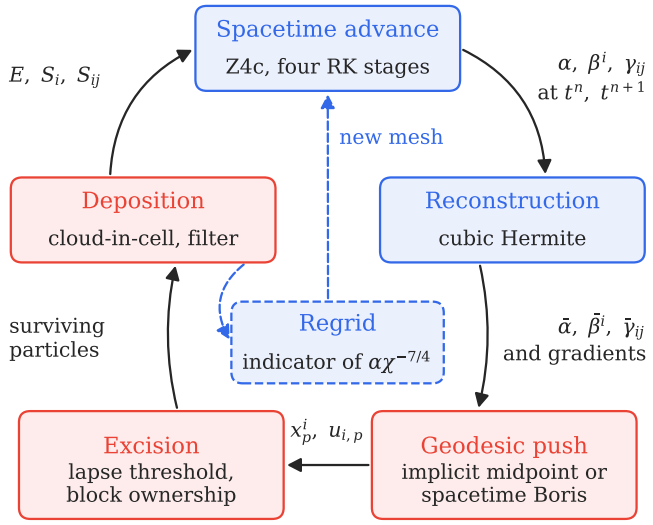


FIG. 1. One step of the particle-mesh cycle from t^n to t^{n+1}. Blue stages act on the mesh, red stages on the macroparticles. Arrows state the quantities passed; overbars mark metric fields averaged over the step. On the dashed regrid branch, metric fields, block ownership, and sources are rebuilt on the new mesh.

and $\bar W = \sqrt{1+\bar\gamma^{ij}\bar u_i\bar u_j}$, advancing the trajectory by the midpoint update,

$$\frac{x^{i,n+1}-x^{i,n}}{\Delta t} = \frac{\bar\alpha\bar\gamma^{ij}\bar u_j}{\bar W} - \bar\beta^i,$$
$$\frac{u_i^{n+1}-u_i^n}{\Delta t} = -\bar W\partial_i\bar\alpha + \bar u_j\partial_i\bar\beta^j + \frac{\bar\alpha}{2\bar W}\bar u^j\bar u^k\partial_i\bar\gamma_{jk}, \quad (9)$$

where $\bar u^j\bar u^k\partial_i\bar\gamma_{jk} = -\bar u_j\bar u_k\partial_i\bar\gamma^{jk}$. Fixed by the coordinate metric alone, the midpoint update conserves any momentum u_i conjugate to a cyclic coordinate ($\partial_i = 0$) to rounding error [27].

The spacetime Boris pusher projects momentum onto the normal-observer tetrad, pairing $e^\mu_{(0)} = n^\mu$ with a spatial triad $e_i^{(m)}$ of three mutually perpendicular unit covectors on the slice [21, 28, 29]. The triad and its inverse $e^i_{(m)}$ satisfy

$$\gamma_{ij} = \delta_{mn}e_i^{(m)}e_j^{(n)}, \qquad e^i_{(m)}e_i^{(n)} = \delta^n_m, \quad (10)$$

with Kronecker deltas δ_{mn} and δ^n_m. Projection defines the tetrad momentum $u_{(m)} = e^i_{(m)}u_i$ (so $u_i = e_i^{(n)}u_{(n)}$) and $W = \sqrt{1+\delta^{mn}u_{(m)}u_{(n)}}$. Along a trajectory with $u^j = \gamma^{jk}u_k$, coordinate velocity v^j, and convective derivative

$$\frac{\mathrm{d}}{\mathrm{d}t} = \partial_t + v^j\partial_j, \qquad v^j = \frac{\alpha u^j}{W} - \beta^j, \quad (11)$$

differentiating $u_i = e_i^{(n)}u_{(n)}$ gives

$$e_i^{(n)}\frac{\mathrm{d}u_{(n)}}{\mathrm{d}t} = \frac{\mathrm{d}u_i}{\mathrm{d}t} - u_{(n)}\big(\partial_t e_i^{(n)} + v^j\partial_j e_i^{(n)}\big). \quad (12)$$

Spatial differentiation of $\gamma_{jk} = \delta_{mn}e_j^{(m)}e_k^{(n)}$ rewrites the metric-gradient term in the characteristic equations as

$$\frac{1}{2}u^ju^k\partial_i\gamma_{jk} = u^ju_{(n)}\partial_i e_j^{(n)}. \quad (13)$$

With the momentum-contracted triad gradients

$$\Omega_{jk} = u_{(n)}\partial_j e_k^{(n)}, \qquad \Omega^t_k = u_{(n)}\partial_t e_k^{(n)}, \quad (14)$$

inserting the characteristic equations, contracting with $e^i_{(m)}$, and passing to normal-observer proper time $\mathrm{d}\hat t = \alpha\,\mathrm{d}t$ yield the tetrad equation of motion,

$$\frac{\mathrm{d}u_{(m)}}{\mathrm{d}\hat t} = \tilde E_{(m)} + \frac{1}{W}\epsilon_{(m)(n)(k)}u_{(n)}\tilde B_{(k)}, \quad (15)$$

where $\epsilon_{(m)(n)(k)}$ is the Levi-Civita symbol. Here the momentum-contracted gravito-electric and gravitomagnetic fields $\tilde E_{(m)}$ and $\tilde B_{(k)}$ read

$$\alpha\tilde E_{(m)} = e^i_{(m)}\big(-W\partial_i\alpha + u_j\partial_i\beta^j + \beta^j\Omega_{ji} - \Omega^t_i\big),$$
$$\tilde B_{(k)} = \epsilon_{(k)(m)(n)}e^i_{(m)}e^j_{(n)}\Omega_{ij}, \quad (16)$$

representing the tetrad field strengths $\partial_\mu e_\nu^{(m)} - \partial_\nu e_\mu^{(m)}$ contracted with momentum [28, 29].

Plasma particle-in-cell codes integrate the Lorentz force with the Boris step [6, 26], a velocity-space counterpart of kick-drift-kick leapfrog that brackets a magnetic rotation at constant W between two half-step electric kicks. In the tetrad equation of motion, $\tilde E_{(m)}$ carries the lapse, shift, and temporal triad gradients that change W, whereas $\tilde B_{(k)}$ carries the spatial triad curl and rotates $u_{(m)}$ at fixed W. Both fields depend on momentum, making the update implicit. Each fixed-point iteration forms $u^n_{(m)} = e^i_{(m)}(t^n, \boldsymbol{x}^n)u^n_i$ and $u^{n+1}_{(m)} = e^i_{(m)}(t^{n+1}, \boldsymbol{x}^{n+1})u^{n+1}_i$ and evaluates $\bar{\tilde E}_{(m)}$, $\bar{\tilde B}_{(k)}$, and $\bar W$ at $\bar{\boldsymbol{x}}$ from their mean $\bar u_{(m)}$. With $\Delta\hat t = \bar\alpha\Delta t$ and $\boldsymbol{u}_{\mathrm{tet}} = (u_{(1)}, u_{(2)}, u_{(3)})$, momentum and position advance through the Boris rotation,

$$\boldsymbol{u}^- = \boldsymbol{u}^n_{\mathrm{tet}} + \frac{\Delta\hat t}{2}\bar{\tilde{\boldsymbol{E}}},$$
$$\boldsymbol{\vartheta} = \frac{\Delta\hat t}{2\bar W}\bar{\tilde{\boldsymbol{B}}},$$
$$\boldsymbol{u}' = \boldsymbol{u}^- + \boldsymbol{u}^-\times\boldsymbol{\vartheta},$$
$$\boldsymbol{u}^+ = \boldsymbol{u}^- + \frac{2(\boldsymbol{u}'\times\boldsymbol{\vartheta})}{1+|\boldsymbol{\vartheta}|^2},$$
$$\boldsymbol{u}^{n+1}_{\mathrm{tet}} = \boldsymbol{u}^+ + \frac{\Delta\hat t}{2}\bar{\tilde{\boldsymbol{E}}},$$
$$\frac{x^{i,n+1}-x^{i,n}}{\Delta t} = \frac{\bar\alpha\bar e^i_{(m)}\bar u_{(m)}}{\bar W} - \bar\beta^i. \quad (17)$$

After convergence, $u^{n+1}_{(m)}$ converts back to $u^{n+1}_i = e_i^{(m)}(t^{n+1}, \boldsymbol{x}^{n+1})u^{n+1}_{(m)}$.

In flat polar coordinates, a free particle follows a straight line while the local radial and azimuthal unit vectors rotate along its path. A discrete step that projects

momentum onto those unit vectors at the endpoints and evaluates their rotation at the midpoint makes an error absent in Cartesian axes. The spatial triad $e_i^{(m)}$ is such a local frame, fixed by γ_{ij} only up to a point-dependent rotation. In the Boris rotation, $e^i_{(m)}$ and $e_i^{(m)}$ project momentum at $(t^n, \boldsymbol{x}^n)$ and $(t^{n+1}, \boldsymbol{x}^{n+1})$, while $\bar{\bar{E}}_{(m)}$ and $\bar{\bar{B}}_{(k)}$ evaluate the triad gradients Ω_{jk} and Ω^t_k at the midpoint. Along a continuous geodesic, those gradients in the tetrad equation of motion cancel the endpoint projections identically. Over a finite step Δt, midpoint evaluation leaves a second-order truncation error set by this local frame choice, the tetrad gauge. Two gauges are implemented here. The symmetric gauge uses the symmetric positive-definite square root of γ_{ij},

$$e_j^{(m)} = \left(\gamma^{1/2}\right)_{mj}, \tag{18}$$

from eigendecomposition, with spatial derivatives solved from the Sylvester equation for $\partial(\gamma^{1/2})$ in the eigenbasis. The Cholesky gauge uses the lower-triangular factor L_{jm} satisfying $\gamma_{ij} = \delta^{mn} L_{im} L_{jn}$,

$$e_j^{(m)} = L_{jm}, \qquad L_{jm} = 0 \text{ for } j < m, \tag{19}$$

equivalent to Gram–Schmidt orthonormalization of the coordinate basis in index order in closed form at lower cost. Reference [21] leaves its tetrad gauge unstated. On its geodesic benchmarks (Sec. IV), the symmetric gauge reproduces all published digits of the invariant errors for analytic spherical Kerr–Schild input. On the reconstructed Cartesian mesh, the Cholesky gauge in (x, y, z) order reproduces all published digits. Permuted coordinate orderings and cross-paired gauges give distinct errors. Because these benchmarks sample the triad only along the tested trajectories, they fix the unstated choice up to any rule that reduces to those two frames there.

B. Metric reconstruction

Geodesic integration on a mesh requires metric components and first spatial derivatives between cell centers with continuous values and gradients across cell faces. The mesh stores α, β^i, and the six covariant components γ_{ij} at cell centers, and the interpolated 3×3 spatial metric is inverted afterward to preserve symmetry.

Along each axis, a four-point cubic Hermite (Catmull–Rom) stencil spans two cell centers on each side of the bracketing interval at fractional offset $s \in [0, 1)$ in units of the cell spacing h. Interpolation and differentiation evaluate the Hermite weights,

$$\begin{aligned} w_{-1}(s) &= \frac{-s + 2s^2 - s^3}{2}, \\ w_0(s) &= \frac{2 - 5s^2 + 3s^3}{2}, \\ w_1(s) &= \frac{s + 4s^2 - 3s^3}{2}, \\ w_2(s) &= \frac{-s^2 + s^3}{2}, \\ w'_a(s) &= \frac{1}{h}\frac{\mathrm{d}w_a}{\mathrm{d}s}. \end{aligned} \tag{20}$$

Tensor products $w_a(s_1) w_b(s_2) w_c(s_3)$ over the 4^3 surrounding cells reconstruct each field, while substituting $w'_a(s_i)$ along axis i gives the exact polynomial derivative ∂_i. The stencil requires two ghost cells per mesh block, as allocated in fixed-spacetime mesh tests, whereas evolved-spacetime runs allocate four. Each macroparticle interpolates within its owning block, shifting the stencil inward by one cell to extrapolate the same cubic when an iterate nears the outer edge of the allocated cells. For standalone grid tests, the same routine samples an analytic metric on a virtual uniform lattice of cell centers of spacing h.

C. Stress-energy deposition

Einstein's equations require smooth grid stress-energy components whose proper-volume integrals equal the discrete particle sums. Each macroparticle carries a shape function, a localized profile of unit integral that sets its cloud width on the grid [6, 22]. On a uniform lattice of spacing Δx and cell centers $\boldsymbol{x}_q$, tensor products of the first-order (cloud-in-cell) shape function B_1 define eight nonvanishing lattice weights $\tilde{\omega}_{qp}$,

$$\begin{aligned} B_1(\xi) &= \max\bigl(0,\, 1 - |\xi|\bigr), \\ \tilde{\omega}_{qp} &= \prod_{i=1}^{3} B_1\left(\frac{x^i_q - x^i_p}{\Delta x}\right), \end{aligned} \tag{21}$$

which are nonnegative and sum to unity. Division of the accumulated moments in leaf cell C by its proper spatial volume $\sqrt{\gamma_C}\,\Delta V_C$, with cell-centered determinant γ_C and coordinate volume ΔV_C, yields the ten deposited sources,

$$\begin{aligned} \bigl(E,\, S_i,\, S_{ij}\bigr)_C = {} & \frac{1}{\sqrt{\gamma_C}\,\Delta V_C} \\ & \times \sum_{p=1}^{N} \omega_{Cp}\, m_p \left(W_p,\, u_{i,p},\, \frac{u_{i,p} u_{j,p}}{W_p}\right), \end{aligned} \tag{22}$$

with dimensionless leaf-cell weight ω_{Cp}.

At a coarse-fine mesh boundary, evaluating B_1 separately on each cell with its own width Δx_C pairs unequal cloud widths on the two sides. For a straddling

particle, those weights fail to sum to unity, so mesh integrals differ from the particle sums. A single representation level ℓ_p for each macroparticle p restores unit total weight [21, 30]. Its cloud lives on a virtual uniform lattice of spacing $\Delta x(\ell_p)$, producing eight unit-sum weights $\tilde{\omega}_{qp}$ on virtual cells q of volume ΔV_q (Fig. 2). Those virtual weights map onto leaf cells C of level ℓ_C and volume ΔV_C through

$$\omega_{Cp} = \begin{cases} \sum_{q \subset C} \tilde{\omega}_{qp}, & \ell_C \le \ell_p, \\ \tilde{\omega}_{qp} \dfrac{\Delta V_C}{\Delta V_q}, & C \subset q,\ \ell_C > \ell_p. \end{cases} \tag{23}$$

Coarser leaf cells ($\ell_C \le \ell_p$) sum enclosed virtual weights, while finer leaf cells ($\ell_C > \ell_p$) share the virtual cell's coordinate density in proportion to volume. At the cell spacing of particle p's own block, eight lattice cell centers bracket $\boldsymbol{x}_p$, and $\ell_{\mathrm{loc}}(p)$ is the finest level among the leaf blocks containing them. The choice $\ell_p = \ell_{\mathrm{loc}}(p)$ defines finest-level clouds [21, 30], where only the $\ell_C \le \ell_p$ branch occurs. Every production run here uses finest-level clouds, while level-capped runs serve as controls (Table I). Because $\sum_C \omega_{Cp} = 1$ for every interior cloud, proper-volume-weighted leaf sums obey the conservation identity,

$$\sum_C \big(E,\, S_i,\, S_{ij}\big)_C \sqrt{\gamma_C}\, \Delta V_C = \sum_{p=1}^{N} m_p \left(W_p,\, u_{i,p},\, \frac{u_{i,p} u_{j,p}}{W_p}\right), \tag{24}$$

to rounding error across arbitrary refinement boundaries. For particles within half a cloud cell of the outer boundary, weight outside the mesh is dropped. Enclosing blocks are found by geometric search, and cell updates accumulate via concurrent atomic additions.

Each refinement level halves Δx and divides N_{ppc} by eight, leaving fine cells under diffuse matter with far less than one particle on average. An optional level-capped deposition bounds ℓ_p through the level cap,

$$\ell_p = \min\big(\ell_{\mathrm{loc}}(p),\, \ell_{\mathrm{cap}}\big), \tag{25}$$

with a density-adaptive ceiling ℓ_{cap}, and recovers finest-level clouds as $\ell_{\mathrm{cap}} \to \infty$ (Fig. 2). In control runs, ℓ_{cap} is the finest level above the initial matter-grid floor whose cell holds at least one macroparticle in the previous deposit, within the coarsening limit of Table I. Both branches of the leaf-weight mapping preserve $\sum_C \omega_{Cp} = 1$, so the conservation identity still holds to rounding error. Coarser clouds raise constraint residuals near forming black holes and shift horizon masses by a few tenths of a percent, more on lattices, yet bound refinement on randomly sampled matter (Sec. V A). Every production collapse run in Sec. V uses finest-level clouds, while level-capped runs serve as controls.

Short-wavelength sampling fluctuations in (E, S_i, S_{ij}) are smoothed after deposition by n_{f} passes of a separable binomial filter with one-dimensional weights $(1/4, 1/2, 1/4)$ [6, 21]. Ghost cells are filled before each pass by copying at equal level and by restriction and prolongation across refinement levels. Along each axis at wavenumber k, n_{f} binomial passes multiply Fourier modes by $\cos^{2n_{\mathrm{f}}}(k\Delta x/2)$ and add $n_{\mathrm{f}}\Delta x^2/2$ to the second spatial moment of every cloud. They widen every cloud, which acts as a gravitational softening, a short-range smoothing of the gravitational pull over the filter scale [2]. For a self-gravitating equilibrium, this widening makes the deposited source less concentrated than the matter and lowers the proper-volume integral of E slightly across nonuniform $\sqrt{\gamma}$ (Sec. V).

Plasma particle-in-cell codes cancel this second-moment widening by following the n_{f} binomial passes with one separable compensation pass [6] of one-dimensional weights

$$\big(-c,\, 1+2c,\, -c\big), \qquad c = \frac{n_{\mathrm{f}}}{4}. \tag{26}$$

With $s = \sin^2(k\Delta x/2)$, the combined one-dimensional wavenumber response is

$$\widehat{F}(k) = (1-s)^{n_{\mathrm{f}}}(1 + n_{\mathrm{f}} s) = 1 - \mathcal{O}\big((k\Delta x)^4\big). \tag{27}$$

This response equals unity to fourth order in $k\Delta x$, falls monotonically to zero at the Nyquist wavenumber $k = \pi/\Delta x$, cancels the second moment, and carries small negative sidelobes. Reference [21] defines one binomial pass and leaves the pass count of each test unstated. Table I lists each scenario's filter: vacuum runs omit filtering, lattice dust uses one plain pass, test-particle clusters use two, and prolate and self-gravitating cluster runs use four with compensation. With uniform $\sqrt{\gamma}$, both stencils conserve total weight on a single level, whereas mesh-level filtering across coarse-fine ghost zones is nonconservative once particles cross refinement boundaries (Sec. IV).

D. Spacetime evolution

Gravitational collapse and black hole mergers require a hyperbolic spacetime formulation that propagates curvature waves and damps constraint violations. We adopt AthenaK's solver [35, 36] for the Z4c formulation [31, 34, 41] in moving-puncture gauge, the standard coordinate choice for evolving black holes on a fixed Cartesian mesh without excising grid cells [32, 33, 42, 43]. With $K = \gamma^{ij} K_{ij}$, Z4c evolves the conformal variables

$$\begin{aligned} \chi &= \gamma^{-1/3}, & \tilde{\gamma}_{ij} &= \chi \gamma_{ij}, \\ \tilde{A}_{ij} &= \chi\left(K_{ij} - \frac{1}{3}\gamma_{ij} K\right), & \hat{K} &= K - 2\Theta, \end{aligned} \tag{28}$$

alongside the contracted conformal connection $\tilde{\Gamma}^i$ and constraint-propagation scalar Θ. Constraint-damping terms control the Hamiltonian and momentum residuals $\mathcal{H}$ and $\mathcal{M}_i$ with κ_1 in Table I and $\kappa_2 = 0$. Moving-puncture gauge combines 1+log lapse slicing [42, 44] with

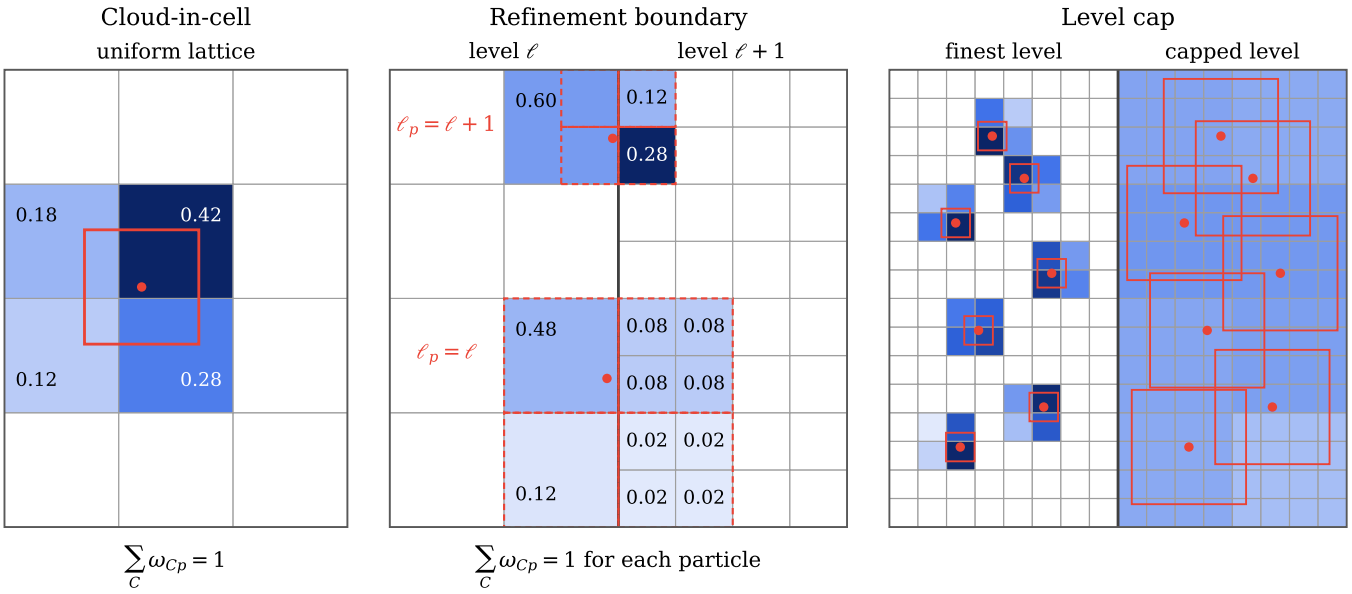


FIG. 2. Single-level stress-energy deposition preserves unit particle weight across refinement boundaries in two dimensions, where each refinement level divides particles per cell by four (eight in three dimensions). Shading marks deposited density, and numbers in the left and middle panels give leaf-cell weights ω_{Cp}. Left: a cloud-in-cell profile (red square) shares one macroparticle among four uniform cells. Middle: virtual weights (dashed cells) sum on a coarser leaf under finest-level clouds ($\ell_p = \ell+1$) and divide uniformly across finer leaves under a level cap ($\ell_p = \ell$). Right: sparse particles deposited with finest-level clouds and with a coarser level cap.

the integrated Gamma-driver shift condition [32, 33, 42] damped at rate η with shift speed parameter μ_S,

$$\partial_t \alpha = \beta^i \partial_i \alpha - 2\alpha \hat{K},$$
$$\partial_t \beta^i = \mu_S \tilde{\Gamma}^i - \eta \beta^i + \beta^j \partial_j \beta^i. \tag{29}$$

As in AthenaK [36], the shift driver uses $\mu_S \tilde{\Gamma}^i$ directly, whereas Ref. [21] writes $\alpha^2 \mu_S \tilde{\Gamma}^i$. Every Z4c run here evolves on a refined mesh with four ghost cells, sixth-order centered finite differences, Kreiss–Oliger dissipation, fourth-order Runge–Kutta integration, and Sommerfeld radiative conditions on outflow outer boundaries [36].

AthenaK's Z4c module, previously coupled solely to magnetohydrodynamics [36], is extended here to receive the particle-deposited sources (E, S_i, S_{ij}) in the evolution equations and constraint residuals $\mathcal{H}$ and $\mathcal{M}_i$. An optional parabolic relaxation of the Hamiltonian constraint [21, 45, 46] adds diffusive damping to the conformal factor evolution,

$$\partial_t \chi = (\partial_t \chi)_{\text{Z4c}} + \frac{c_H}{2} \chi^{-1/4} \mathcal{H},$$
$$\mathcal{H} = {}^{(3)}R + \frac{2}{3} K^2 - \tilde{A}_{ij} \tilde{A}^{ij} - 16\pi E, \tag{30}$$

where $(\partial_t \chi)_{\text{Z4c}}$ is the standard Z4c evolution term and ${}^{(3)}R$ is the Ricci scalar of γ_{ij}. Here the relaxation length c_H is constant in the control run that enables it (Table I), while the code also supports a Δx-proportional term. The Hamiltonian constraint norm $\|\mathcal{H}\|$ is the square root of the proper-volume integral of $\mathcal{H}^2$ over leaf cells, excluding cells with $\chi < 1/16$ and without dividing by volume. Reference [21] instead reports a coordinate-volume root-mean-square average.

E. Adaptive mesh

Gravitational collapse and black hole mergers concentrate steep curvature gradients inside volumes far smaller than the ambient cluster or wave zone. AthenaK uses block-structured octree refinement, splitting cubic mesh blocks recursively into eight half-sized child blocks where finer resolution is required [35, 47]. Uniform and statically refined meshes enter as the limits without dynamic block splitting.

Dynamic refinement tracks the scalar monitor

$$q = \alpha \chi^{-7/4}, \tag{31}$$

which rises sharply in strong-field regions as $\chi^{-7/4}$ grows near black holes [21]. With $\boldsymbol{\iota}_a$ the unit cell shift along axis $a \in \{1, 2, 3\}$ and $\Delta_a q_i = q_{i+\boldsymbol{\iota}_a} - q_{i-\boldsymbol{\iota}_a}$, each cell i evaluates a Löhner-type second-difference ratio [21, 48],

the refinement indicator,

$$\varepsilon_i^2 = \frac{\sum_{a,b=1}^{3}\left(\Delta_a q_{i+\iota_b} - \Delta_a q_{i-\iota_b}\right)^2}{\sum_{a,b=1}^{3}\left(|\Delta_a q_{i+\iota_b}| + |\Delta_a q_{i-\iota_b}| + \epsilon_0 q_{ab,i} + \delta_0\right)^2},$$
$$q_{ab,i} = \sum_{\sigma,\tau=\pm1} \left|q_{i+\sigma\iota_a+\tau\iota_b}\right|. \tag{32}$$

As in Ref. [21], the filter weight is $\epsilon_0 = 0.01$ and the positive floor is $\delta_0 = 10^{-99}$. Blocks refine above $\max_i \varepsilon_i = 0.2$ and derefine below 0.02, checked every four cycles with regrids at least eight cycles apart. Prescribed inner radii or tracked-puncture boxes enforce minimum levels, with mesh spacings listed in Table I and refinement thresholds, regions, and block limits in Table II. Whenever a finer level activates, the global step Δt halves. Through every regrid, macroparticles keep $(m_p, x_p^i, u_{i,p})$ unchanged without splitting or merging, and metric arrays, block ownership, and deposited sources are rebuilt on the new leaf cells.

F. Numerical methods

End-to-end evolutions from initial equilibria through black hole formation rely on interior particle excision, horizon and mass diagnostics, constraint-satisfying initial data, and portable parallel execution.

Inside a black hole, $1+\log$ slicing drives α toward zero while signals remain causally trapped behind the horizon. Lapse excision is the default of Ref. [21] and removes particles with $\alpha(t^n, \boldsymbol{x}_p^n) < \alpha_{\rm exc}$ [49], with $\alpha_{\rm exc}$ listed in Table I. Reference [21] also offers apparent-horizon-based excision and reports no significant difference in its benchmarks. Here, with three to ten cells per horizon radius, the lapse at cell centers next to the puncture stays at or above $\alpha_{\rm exc}$. Lapse excision alone then leaves matter inside the black hole. For the small black holes of Sec. V, particles inside 0.8 of the smallest coordinate radius of a found apparent horizon, measured from its center, are removed in addition. This rule waits for a successful search of the current run and ignores a surface more than four times the starting radius of the search.

Black hole boundaries and irreducible masses $M_{\rm AH}$ are measured on apparent horizons, outermost closed two-surfaces on a slice whose outgoing orthogonal null geodesics have zero expansion. AthenaK's pseudospectral flow finder [36, 50] searches every 16 cycles on 16 polar-angle points up to harmonic degree 8, stopping within 400 iterations once the horizon-mass change drops below 10^{-5}. As distributed, the finder computes finite-difference metric derivatives on the active cells of each mesh block and interpolates that array to the trial surface with unfilled block-face ghost zones. Here metric derivatives on the trial surface come directly from differentiating the eight-point Lagrange interpolant of the ghost-filled metric. Its effect on a black hole of known mass appears in Sec. V. With the default flow constant 1, sized for a unit-mass black hole, the first step for mass $m \ll M$ overshoots the starting radius and ends without a horizon. Searches there set the flow constant to 0.05 instead of 1 for individual horizons of mass near $0.1\,M$ and 0.2 for the common horizon, with stopping tolerance 10^{-7} over 2000 iterations. With those settings, the finder and the Misner–Sharp mass $M_{\rm MS}$ below agree to one percent.

An independent check is provided by the Misner–Sharp mass, the gravitational mass inside a coordinate sphere computed from the slice geometry alone,

$$M_{\rm MS} = \frac{R}{2}\left(1 - \gamma^{ij}\partial_i R\,\partial_j R + R^2 K_T^2\right). \tag{33}$$

Here R is the areal radius of a coordinate sphere of proper area $4\pi R^2$ and K_T is the tangential component of K_{ij} on the sphere. In spherical symmetry, $2M_{\rm MS}/R = 1$ locates the apparent horizon and $M_{\rm MS}$ equals the black hole mass in vacuum. Values of $M_{\rm MS}$ are evaluated on saved slices independently of the finder. For origin-centered runs, central lapse $\alpha_c(t)$ and central energy density $E_c(t)$ average the eight central leaf cells, defining the central normal-observer proper time,

$$\tau_c(t) = \int_0^t \alpha_c(t')\,\mathrm{d}t'. \tag{34}$$

Coordinate-independent curvature is tracked via the Kretschmann scalar $I = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ of the Riemann tensor $R_{\mu\nu\rho\sigma}$, decomposed through the Gauss, Codazzi, and Ricci equations with Einstein's equations for the two-normal projection [21],

$$\begin{aligned} I &= G_{ijkl}G^{ijkl} - 4C_{ijk}C^{ijk} + 4Z_{ij}Z^{ij},\\ G_{ijkl} &= {}^{(3)}R_{ijkl} + K_{ik}K_{jl} - K_{il}K_{jk},\\ C_{ijk} &= D_iK_{jk} - D_jK_{ik},\\ Z_{ij} &= {}^{(3)}R_{ij} + KK_{ij} - K_{ik}K^k{}_j\\ &\quad - 8\pi\left[S_{ij} - \frac{1}{2}\gamma_{ij}(S-E)\right]. \end{aligned} \tag{35}$$

Here ${}^{(3)}R_{ijkl}$ and ${}^{(3)}R_{ij}$ are the Riemann and Ricci tensors of γ_{ij}, $S = \gamma^{ij}S_{ij}$, and indices are raised with γ^{ij}. Spatial derivatives in I use sixth-order centered finite differences, checked against the analytic Schwarzschild profile.

Self-gravitating runs start on conformally flat, time-symmetric slices—surfaces of instantaneous rest with $K_{ij} = 0$, $\beta^i = 0$, and $\gamma_{ij} = \psi^4\delta_{ij}$ for conformal factor ψ—where $\mathcal{M}_i = 0$ identically. Linear superposition of flat-space Poisson potentials u_c for conformal density sources $\tilde{\rho}_c(\boldsymbol{x})$ solves $\mathcal{H} = 0$ via

$$\begin{aligned} \psi(\boldsymbol{x}) &= 1 + \sum_c u_c(\boldsymbol{x}),\\ \nabla^2_{\rm flat}u_c &= -2\pi\tilde{\rho}_c,\\ E(\boldsymbol{x}) &= \psi(\boldsymbol{x})^{-5}\sum_c \tilde{\rho}_c(\boldsymbol{x}). \end{aligned} \tag{36}$$

Components combine tabulated spherical equilibria, Gaussians, closed-form homogeneous prolate spheroids, and vacuum black hole punctures

$$u_c(\boldsymbol{x}) = \frac{m_c}{2\bar{r}_c} \tag{37}$$

of bare mass m_c at isotropic distance $\bar{r}_c$. With $\sqrt{\gamma} = \psi^6$, equal-mass ensembles of rest-mass density ρ_0 have coordinate number density proportional to $\rho_0\psi^6$. Cold spherical collapse places macroparticles on a mesh-aligned cubic lattice of spacing h, where published lattice masses [21] set $m_p = \rho_0(\boldsymbol{x}_p)\psi(\boldsymbol{x}_p)^6 h^3$ inside the surface. The number of lattice sites inside a sphere differs from its volume in cells by a surface fraction inversely proportional to the cell count across, giving a rest-mass offset at low N. Here a surface correction rescales m_p in the outermost layer of width h by a common factor so $\sum_p m_p$ equals the continuum rest mass M_0. Interior masses stay unchanged, and omitting this rescaling recovers published lattice masses (Sec. V). Random-sample dust controls draw equal-mass particles ($m_p = M_0/N$) uniformly inside the coordinate ball with acceptance probability $\psi(\boldsymbol{x}_p)^6/\psi(0)^6$ [21]. Spherical clusters draw particles by randomized stratified sampling in radius and in both position and velocity angles, or by independent Monte Carlo draws in sampling comparisons. The mass-weighted mean of $u_{i,p}$ is subtracted so net cluster momentum vanishes, while prolate spheroids use equal-mass Monte Carlo sampling with reflection and half-turn symmetries.

All mesh and particle kernels run through Kokkos [35, 37] across multicore processors and graphics accelerators, with particle updates taking a small fraction of the spacetime step cost at $N \leq 10^4$. Optional features of Ref. [21] omitted here comprise the auxiliary-field shift condition, tracked-puncture profiles of η and κ_1, its second horizon-finder library, higher-degree shape functions, and electromagnetic forces in the pusher.

IV. TEST PARTICLES

Passive trajectories in a known spacetime isolate geodesic integration, metric reconstruction, deposition, and interior excision from gravitational back-reaction. In this test-particle regime, macroparticles follow timelike geodesics of a fixed or independently evolved vacuum metric while deposited sources (E, S_i, S_{ij}) serve as passive diagnostics. Direct comparison with exact geodesics and published benchmarks [21] tests every particle stage in units of the black hole ADM mass M.

A. Fixed black hole

Bound orbits around a spinning black hole test whether both pushers preserve relativistic integrability and precession over many dynamical periods. Around a Kerr black hole of spin a in Boyer–Lindquist or spherical Kerr–Schild coordinates (t, r, θ, ϕ), timelike geodesics conserve the Killing energy $\mathcal{E}$, axial angular momentum L_z, and Carter constant Q,

$$\mathcal{E} = -u_t, \qquad L_z = u_\phi,$$
$$Q = u_\theta^2 + \cos^2\theta \left[a^2\left(1 - \mathcal{E}^2\right) + \frac{L_z^2}{\sin^2\theta}\right]. \tag{38}$$

In the nonrotating limit ($a \to 0$), spherical symmetry reduces Q to $L_{\mathrm{orb}}^2 - L_z^2$ for total orbital angular momentum L_{orb}. At spin $a = 0.9\,M$, equatorial eccentric and inclined Boyer–Lindquist orbits stay on their invariant tori over many radial periods (Fig. 3). Both pushers match exact apsidal and nodal precession with bounded invariant errors and convergent fixed-point iterations. Because the Boyer–Lindquist spatial metric is diagonal, both tetrad gauges coincide there, and the midpoint update conserves cyclic L_z to rounding error [27].

A near-capture eccentric orbit probes strong-field accuracy against the benchmarks of Ref. [21]. In Kerr–Schild coordinates, an inclined trajectory ($a = 0.9\,M$, $e = 0.99$) passes close to the event horizon before returning outward (Fig. 4). Because the midpoint update of Sec. III A depends solely on the coordinate metric and its spatial derivatives, its discrete trajectory is uniquely fixed. On the analytic spherical Kerr–Schild metric, the implicit midpoint pusher conserves cyclic L_z to rounding error and matches all published digits of the maximum relative errors in $\mathcal{E}$ and Q [21]. On the Cartesian Kerr–Schild metric reconstructed with the Hermite weights of Sec. III B, where the mesh breaks continuous axisymmetry, it again matches all published digits for $\mathcal{E}$, L_z, and Q [21].

Off-diagonal spatial metric terms in Kerr–Schild coordinates separate the symmetric gauge from the Cholesky gauge in the spacetime Boris pusher. Continuous geodesics are frame independent because triad connections in the tetrad equation of motion cancel endpoint projections identically. Over a finite step Δt, midpoint evaluation of the Boris rotation leaves a gauge-dependent second-order error. On analytic spherical Kerr–Schild input, the symmetric gauge reproduces all published digits of the $\mathcal{E}$, L_z, and Q errors in Ref. [21] (Fig. 4), whereas the (r, θ, ϕ) Cholesky gauge gives distinct values. On the mesh-reconstructed Cartesian metric, the (x, y, z) Cholesky gauge reproduces all published digits, whereas the symmetric gauge and permuted Cholesky orderings give distinct values.

Independent sweeps of Δt and cell spacing Δx measure temporal and spatial convergence across spinning and nonrotating black hole spacetimes. On analytic metrics, position and invariant errors of both pushers and gauges fall as Δt^2 across the bound Kerr orbits and the $e = 0.99$ passage (Fig. 4). In the nonrotating limit ($a \to 0$), circular, inclined eccentric, and plunging orbits are tested against exact Schwarzschild solutions (Fig. 5). Spherical Kerr–Schild circular motion advances linearly in time

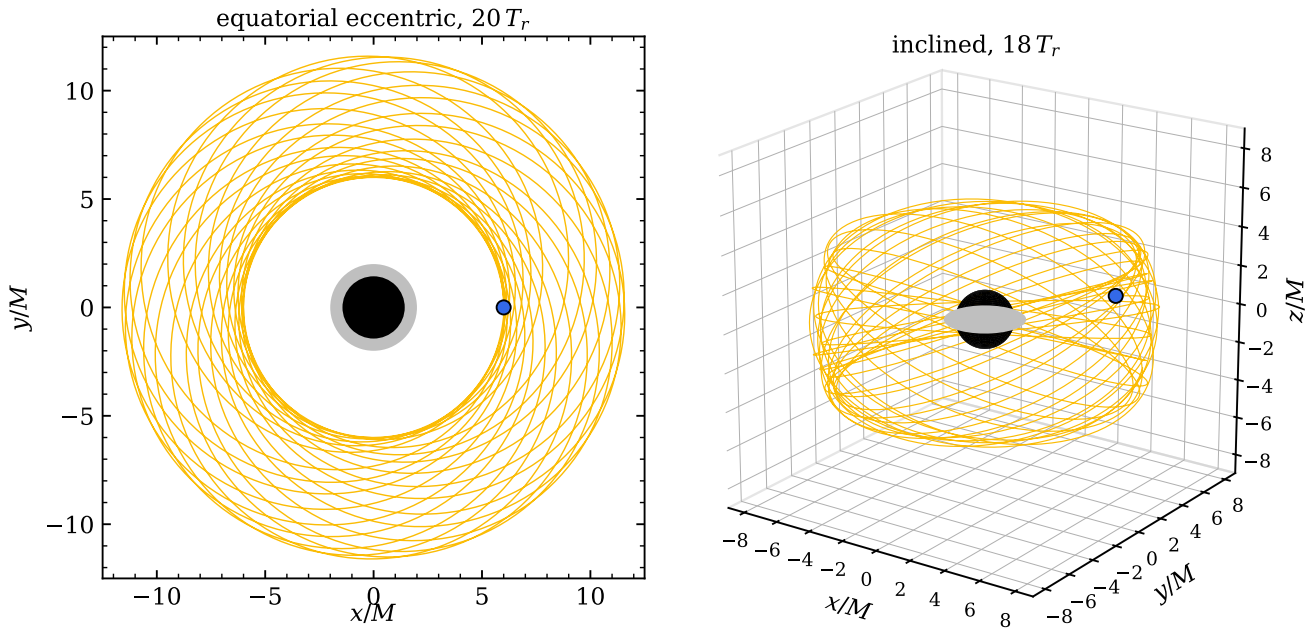


FIG. 3. Bound test-particle trajectories around a Kerr black hole ($a = 0.9\,M$) precess coherently and stay confined to their invariant tori. Left: equatorial eccentric orbit; right: inclined three-dimensional orbit. Solid yellow curves show the spacetime Boris pusher in the symmetric gauge on the analytic Boyer–Lindquist metric from initial blue markers outside the black event horizon and gray ergosphere.

and azimuth and integrates to rounding error. Cartesian circular, eccentric, and plunging orbits converge as Δt^2, and multi-block mesh runs at fine Δt match virtual-lattice runs of spacing Δx to rounding error across block faces. Grid refinement in Δx reduces metric, position, and invariant errors at third order for $a \to 0$ (Fig. 5) and $a = 0.9\,M$ (Table III).

Analytic Cartesian trajectories separate spatial interpolation error from Cartesian time-stepping error on the $e = 0.99$ passage. Reference [21] attributes the mesh error of both pushers on this orbit to spacetime interpolation. Time-step sweeps on the analytic Cartesian metric alongside the Hermite mesh at fixed Δx isolate each contribution. Spatial interpolation sets the $(\mathcal{E}, L_z, Q)$ errors on the mesh at the published step size (Fig. 4, bottom panels). Those mesh invariant errors exceed analytic Cartesian values there and flatten onto an interpolation floor as Δt shrinks. Conversely, the final-position error on the analytic Cartesian metric matches the Hermite mesh closely at the published step size (Fig. 4, top right). Both errors fall as Δt^2. At equal Δt, differences between mesh and analytic-Cartesian final states show that spatial interpolation shifts the position far less than Cartesian time-stepping at the published step size. The two contributions become comparable near $\Delta t = 10^{-3}\,M$.

Orbiting clouds of macroparticles test phase mixing and invariant conservation on a refined mesh. Gaussian clusters of $N = 10^2$, 10^3, and 10^4 bound test particles start around an equatorial circular orbit for $a = 0.9\,M$ and $a \to 0$. Each cluster evolves on both the analytic Cartesian metric and a statically refined octree mesh. Differential orbital shear stretches each cloud into an arc, a trailing spiral, and an azimuthally mixed ring traced by E [2] (Fig. 6). Single-particle $(\mathcal{E}, L_z, Q)$ errors remain bounded independently of N, with mesh L_z and Q errors rising over the first orbit before leveling off (Fig. 6 for $\mathcal{E}$ and L_z). On the mesh, both pushers give matching invariant errors that decrease with Δx at about third order in $\mathcal{E}$ and order 2.5 in L_z. Both pushers also follow the analytic phase-mixing histories closely.

Repeated octree-interface crossings during phase mixing and infall test multi-level source conservation before and after filtering. Every macroparticle deposits (E, S_i, S_{ij}) at the finest local level (Sec. III C). Before filtering, the conservation identity holds to rounding error across all level crossings and excisions whenever clouds sit at least half a cell inside the outer boundary (Fig. 6). For the few wide-orbit particles within half a cell of the outer boundary, the cloud weight outside the mesh accounts to rounding error for the difference. After the filter on the mesh, the integrals match the particle sums before level crossings begin and differ slightly once particles cross refinement boundaries, reflecting the nonconservation of the mesh filter across levels.

Clusters near the capture barrier test lapse excision against exact geodesics and measure sampling-noise scaling with N. Plunging-family clouds ($N = 10^4$) at $a = 0.9\,M$ and $a \to 0$ send many macroparticles across

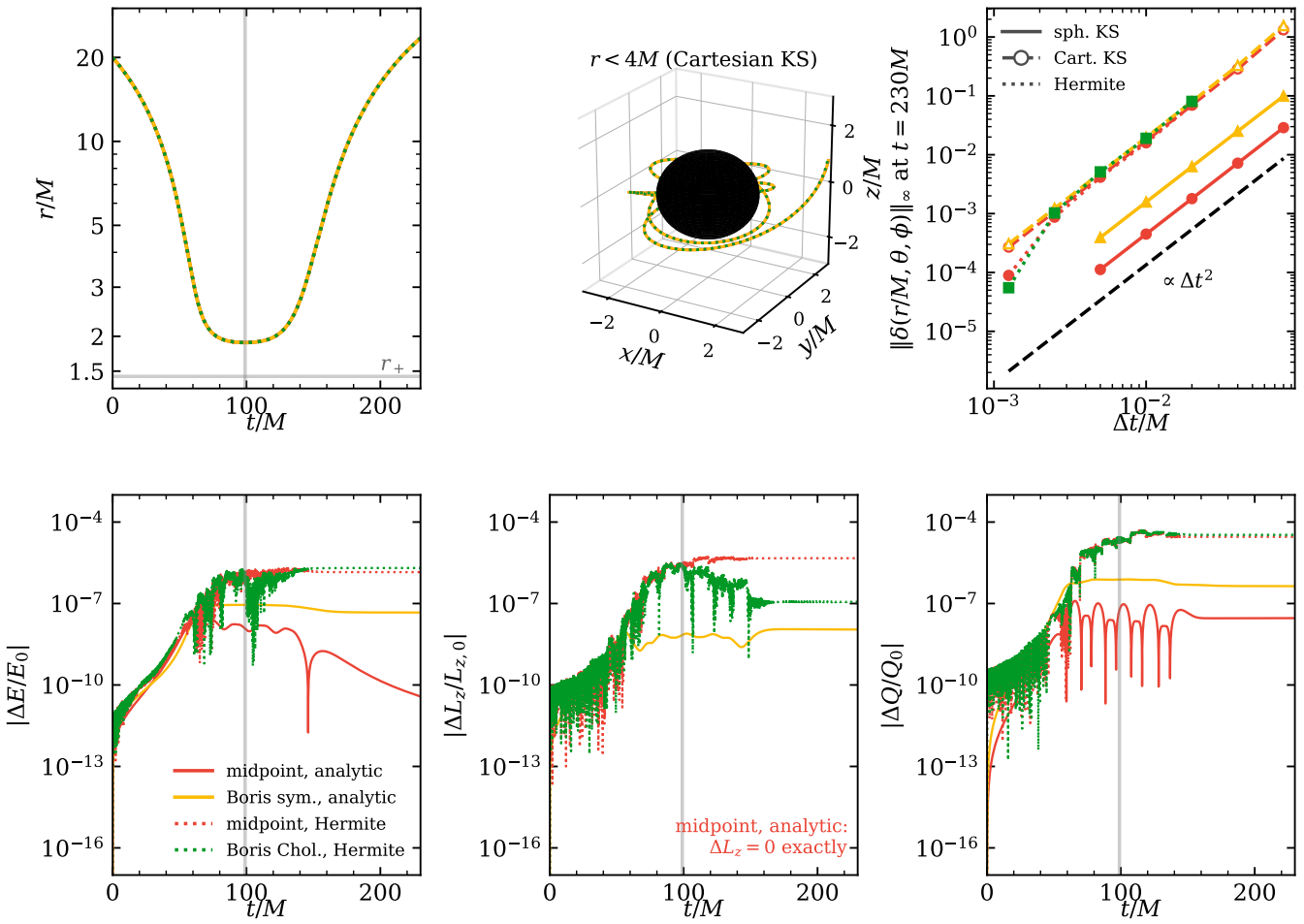


FIG. 4. A near-capture Kerr passage ($a = 0.9\,M$, $e = 0.99$) conserves $(\mathcal{E}, L_z, Q)$ with both pushers and converges as Δt^2. Top: radius against time, trajectory near the black hole, and position error against time step with a second-order guide (black dashed). Bottom: relative errors of $\mathcal{E}$, L_z, and Q against time across periapsis (gray line). Red marks the implicit midpoint pusher, yellow the spacetime Boris pusher in the symmetric gauge, and green the spacetime Boris pusher in the Cholesky gauge. Solid, dashed (top right only), and dotted curves mark analytic spherical Kerr–Schild, analytic Cartesian, and mesh-reconstructed metrics.

the event horizon, where exact Kerr effective potentials predict arrival times at $\alpha_{\rm exc}$. With both pushers on the mesh, lapse excision matches the predicted removal curve up to a few marginal orbits near the capture threshold perturbed by interpolation (Fig. 6). Active, excised, and escaped rest masses sum to the initial cluster mass to rounding error. Across independent bound clusters of $N = 10^2$, 10^3, and 10^4 macroparticles at fixed total mass, pairwise root-mean-square differences in filtered E fall as $N^{-1/2}$.

B. Evolving black holes

Dynamical black hole evolutions in moving-puncture gauge mix physical geodesic motion with coordinate drift. A nonrotating vacuum puncture of initial mass M evolves on a statically refined Cartesian mesh under Z4c (Sec. III D) from time-symmetric data with a precollapsed lapse that starts small near the puncture. Zero-mass tracers advance on the time-averaged Hermite-reconstructed metric with the implicit midpoint pusher. As $1 + \log$ slicing and the Gamma-driver shift relax the slice toward a stationary trumpet [32, 33, 51], coordinates drift outward and $\alpha W - \beta^i u_i$ varies during the gauge transient. Comparison with exact Schwarzschild geodesics therefore uses gauge-independent worldline observables: orbital azimuth, the areal radius R of the coordinate sphere through the particle, and proper time

$$\tau(t) = \int_0^t \frac{\alpha}{W}\,\mathrm{d}t'. \tag{39}$$

Radially infalling and plunging tracers test strong-field worldlines through the evolving apparent horizon to interior excision. A radial infaller from rest and an inspiraling plunger cross refinement boundaries smoothly,

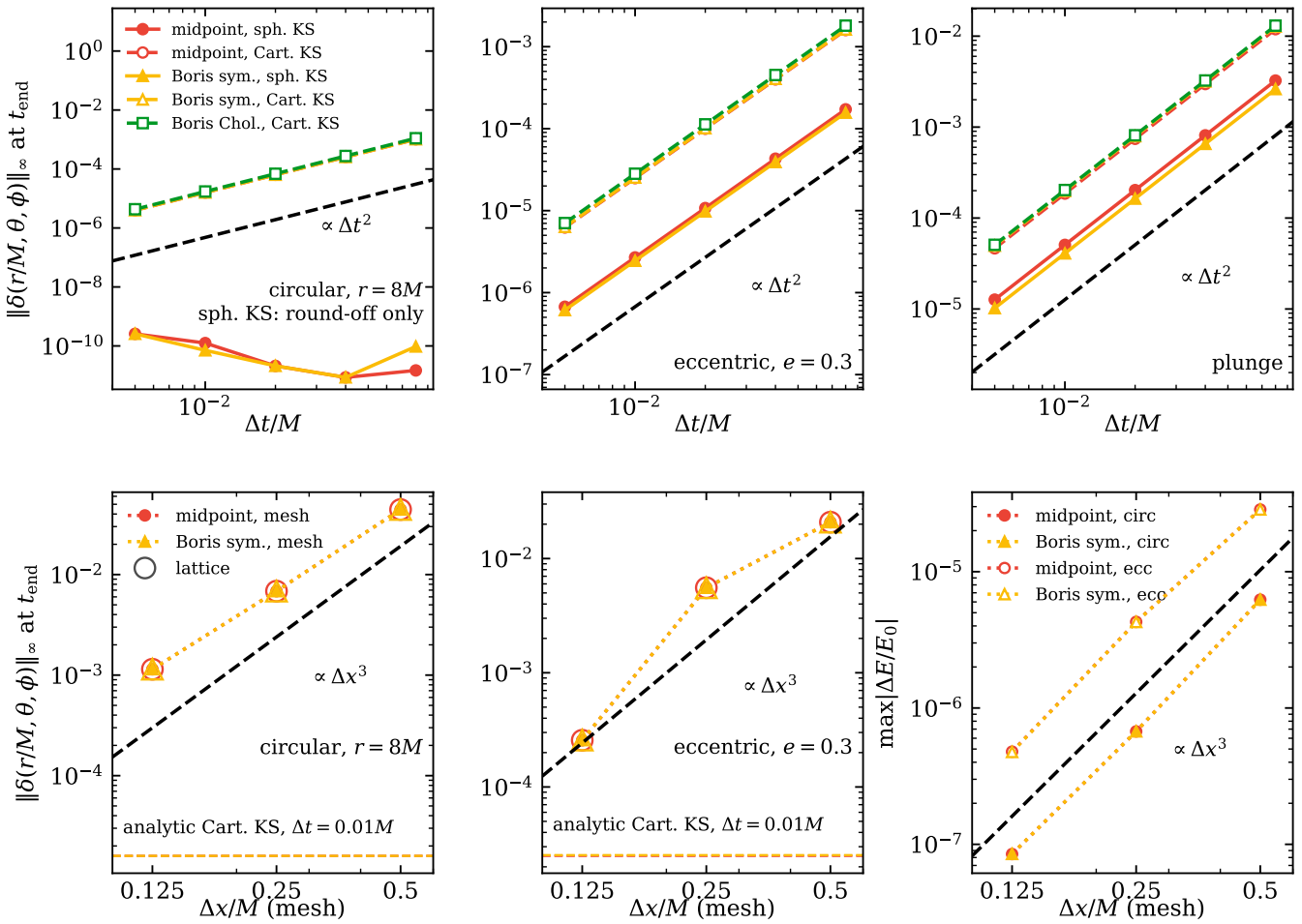


FIG. 5. In the nonrotating limit ($a \to 0$), trajectories converge at second order in Δt on analytic metrics and at third order in cell spacing Δx under cubic Hermite reconstruction. Top: position error versus Δt for circular, eccentric, and plunging orbits; bottom: position and $\mathcal{E}$ errors versus Δx on the mesh (dotted) and virtual lattice (open markers) with power-law guides (black dashed). Curves show the implicit midpoint pusher (red) and the spacetime Boris pusher in the symmetric (yellow) and Cholesky (green) gauges on analytic spherical (solid) and Cartesian (dashed) metrics.

enter the apparent horizon, and undergo lapse excision once the interior lapse drops below $\alpha_{\rm exc}$ (Fig. 7). Along radial infall, $R(\tau)$ tracks the exact Schwarzschild trajectory closely outside the horizon, departing only within a few cells of the puncture where steep gradients limit interpolation of R. Along the slower plunge, $R(\tau)$ and azimuth accumulate a small advance relative to the exact geodesic of mass M. Exact geodesic integration from the post-transient state with the slice Misner–Sharp mass $M_{\rm MS}$ removes nearly that entire offset, as shown for $R(\tau)$ in Fig. 7.

Circular tracers evolved over many periods test whether their small radial oscillations reflect pusher error or the black hole mass in the simulation. Equatorial and inclined circular tracers keep planar orbits and conserved orbital angular momentum $L_{\rm orb}$ while their coordinate radii drift outward and $R(\tau)$ oscillates slightly inside its initial value. Saved slices show that the exterior settles into a stationary Schwarzschild slice whose Misner–Sharp mass $M_{\rm MS}$ slightly exceeds M at this resolution. With the angular momentum of mass M, each tracer orbits that slightly heavier numerical black hole from apocenter on a mildly eccentric geodesic matching $R(\tau)$. In Schwarzschild geometry, R, $L_{\rm orb}$, and proper radial acceleration imply an enclosed mass

$$M_{\rm eff} = \frac{L_{\rm orb}^2/R - R^2\,\mathrm{d}^2R/\mathrm{d}\tau^2}{1 + 3L_{\rm orb}^2/R^2}, \tag{40}$$

which agrees across all orbital radii and inclinations with $M_{\rm MS}$ of the slice (Fig. 7).

A head-on collision of two equal-mass black holes starting from rest extends these tracer tests to a dynamical merger with adaptive mesh refinement following both punctures. Axisymmetry about the collision axis makes each particle's angular momentum about that axis the one exact geodesic invariant of the spacetime. Orbiting

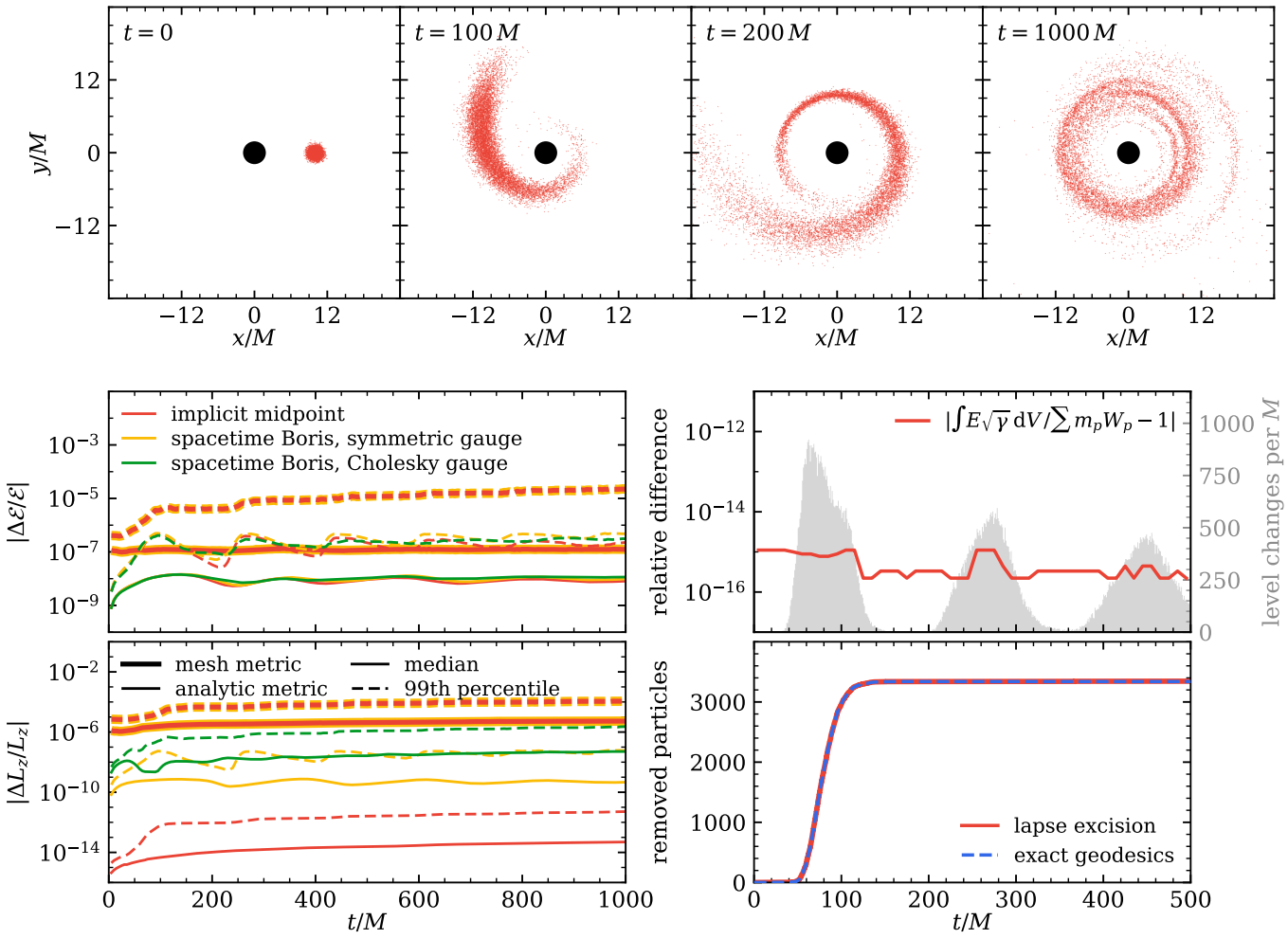


FIG. 6. Clusters of 10^4 test particles around a Kerr black hole ($a = 0.9\,M$). Top: a circular-orbit cluster phase-mixes into a ring on the mesh (black disk: black hole). Bottom left: median (solid) and 99th-percentile (dashed) relative errors of $\mathcal{E}$ and L_z on the mesh (thick) and analytic metric (thin); red implicit-midpoint curves overlap yellow spacetime-Boris curves. Bottom right, for a plunging-family cluster: unfiltered conservation error (red) and refinement-crossing rate (gray, right axis) above; particles removed by lapse excision (red) and exact geodesics (blue dashed) below.

tracers conserve that angular momentum closely through merger within the axisymmetry truncation error of the Cartesian mesh, while a symmetry-plane tracer from rest holds zero angular momentum to rounding error until excision. Once the punctures meet, the mass of the common apparent horizon exceeds the total mass slightly, measuring the error of the spacetime evolution at this resolution.

V. SELF-GRAVITATING MATTER

Coupled Einstein-Vlasov dynamics begins once macroparticles carry finite rest mass m_p and feed their deposited sources (E, S_i, S_{ij}) back into the spacetime evolution. Collective gravitational wells then form, oscillate, disperse, or collapse into black holes from the matter distribution itself, closing the particle-mesh cycle of Sec. III. Whereas Ref. [21] evolves its main self-gravitating benchmarks with about 10^6 to 5×10^6 macroparticles on large machines, every run here uses at most ten thousand ($N \leq 10^4$) on one graphics card. Quantities independent of N, such as continuum equilibrium sequences and analytic collapse bounds, test the coupled solver at full precision. Phenomena sensitive to discrete sampling—initial lattices, multi-pass filtering, refinement under diffuse matter, and cluster holding—isolate what changes at $N \leq 10^4$, with scenario parameters listed in the appendix.

A. Dust collapse

Spherical collapse of cold, pressureless matter from rest tests the coupled solver from smooth initial curvature through black hole formation against an exact

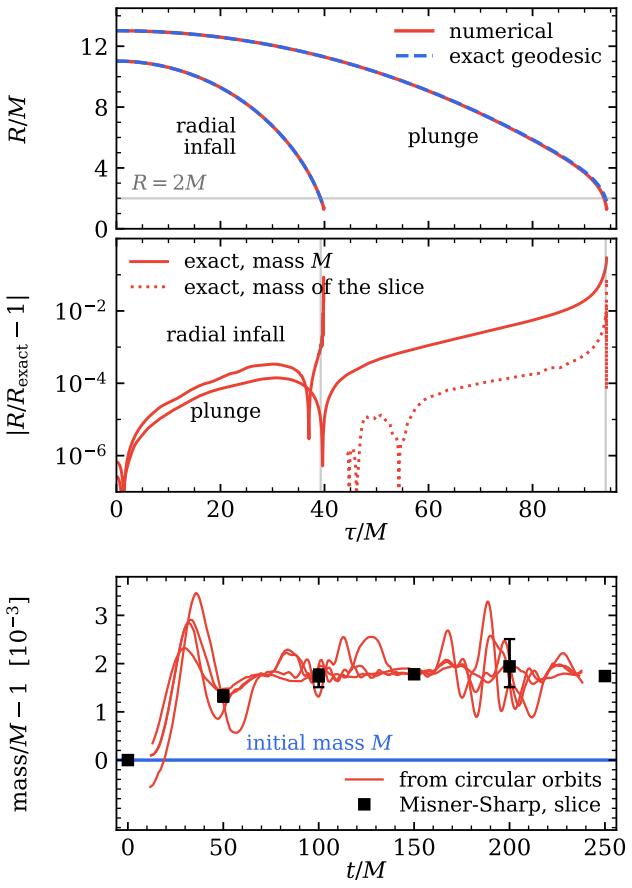


FIG. 7. Test particles around an evolved puncture of mass M. Top: areal radius against proper time for infalling tracers (red) and exact geodesics (blue). Middle: relative difference with mass M (solid) or slice mass $M_{\rm MS}$ (dotted); gray: $R = 2M$. Bottom: circular-orbit mass (red) and $M_{\rm MS}$ (black).

solution [38]. A homogeneous dust ball of initial areal radius $R_0 = 5\,M$, ADM mass M, and uniform density $E_0 = \rho_0 = 3M/(4\pi R_0^3)$ starts at rest on a conformally flat slice [21, 52, 53]. At $t = 0$, momenta $u_{i,p}$, extrinsic curvature K_{ij}, and shift β^i vanish, $\alpha = 1$, and interior and exterior ψ join smoothly at isotropic radius $\bar{r}_0$. Nearly 10^4 macroparticles occupy a cubic lattice of spacing h inside $\bar{r}_0$ and evolve with dynamic refinement, one plain binomial filter pass, and lapse excision. Along free-falling worldlines parameterized by $\varphi \in [0, \pi)$, shell areal radius R, central proper time τ_c, and interior density E follow the cycloid [38],

$$R(\varphi) = \frac{R_0}{2}(1 + \cos\varphi), \qquad \tau_c(\varphi) = \sqrt{\frac{R_0^3}{8M}}\,(\varphi + \sin\varphi),$$
$$E(\varphi) = E_0\left(\frac{R_0}{R(\varphi)}\right)^3. \tag{41}$$

Because $1 + \log$ slicing ties $\partial_t\alpha$ to local volume contraction, α_c tracks central compression directly. From unit initial lapse without shift advection, local integration inside the homogeneous core gives an analytic lower bound on α_c, the lapse bound [53],

$$\alpha_{\rm bound}(\tau_c) = 1 + 6\ln\left(\frac{R(\tau_c)}{R_0}\right). \tag{42}$$

The surface density jump launches an inward gauge wave in the slicing coordinate α. Until that gauge wave reaches the origin, causal isolation keeps $\alpha_c(\tau_c)$ on the lapse bound while the dust falls freely.

Central infall and black hole formation are expected to match continuum Oppenheimer–Snyder collapse despite the discrete particle representation. In the simulation (Fig. 8, top left), $\alpha_c(\tau_c)$ tracks the lapse bound closely during causal isolation and departs smoothly once the surface gauge wave arrives, matching Ref. [21] and continuum runs [53]. Until the center refines to $\Delta x = h/4$, the central density $E_c(\tau_c)$ likewise follows the cycloid closely (Fig. 8, top right, red curve). At $\Delta x = h/4$, fine cells outnumber nearby particles and leave part of the central probe empty, dropping E_c below the cycloid until infall concentrates particles inward. As strong-field curvature grows, $1+\log$ slicing drives α_c toward zero and an apparent horizon forms around the matter (Fig. 8, bottom left). Shortly afterward, interior α drops below $\alpha_{\rm exc}$ and lapse excision removes every macroparticle. Meanwhile, the Hamiltonian constraint norm $\|\mathcal{H}\|$ (Sec. III D; Table IV) peaks and damps across the vacuum ringdown (Fig. 8, bottom right).

At $N \lesssim 10^4$, discrete lattice initialization must match the particle rest-mass sum to the continuum rest mass M_0 of the initial slice. With published lattice masses [21], each site inside $\bar{r}_0$ carries rest mass $m_p = \rho_0\psi(\boldsymbol{x}_p)^6h^3$, the local proper rest-mass density times the cell volume. Those masses were first assumed to yield a black hole of mass M here, as at $N \sim 10^6$ where surface count offsets vanish. At $N \lesssim 10^4$, however, the count of lattice sites inside a sphere differs from its volume in cells. On the lattice of Table I (26 cells across $2\bar{r}_0$), that count exceeds the cell volume and leaves $\sum_p m_p$ above M_0 (Table IV). Runs with published lattice masses kept the interior density exact, so α_c and E_c still matched the surface-corrected collapse. Their apparent-horizon mass $M_{\rm AH}$, however, formed visibly above M (Fig. 8, bottom left, orange curve; Table IV). Radial profiles of the slice Misner–Sharp mass $M_{\rm MS}$ traced this excess to a mass shell leaving the boundary at $t = 0$ at light speed, outside which $M_{\rm MS}$ equaled M. The surface correction of Sec. III F rescales m_p in the outermost lattice layer of width h so $\sum_p m_p = M_0$ with interior masses unchanged. With surface-corrected lattice masses, $M_{\rm AH}$ at formation matches M to within the accuracy of the mass measure and drifts downward only slightly afterward (Fig. 8, bottom left, red curve; Table IV).

Reliable horizon masses after collapse require checking the apparent-horizon finder on a vacuum black hole of known mass M without matter or excision. In initial collapse tests, AthenaK's distributed pseudospectral flow finder [35, 36, 50] returned horizon masses visibly be-

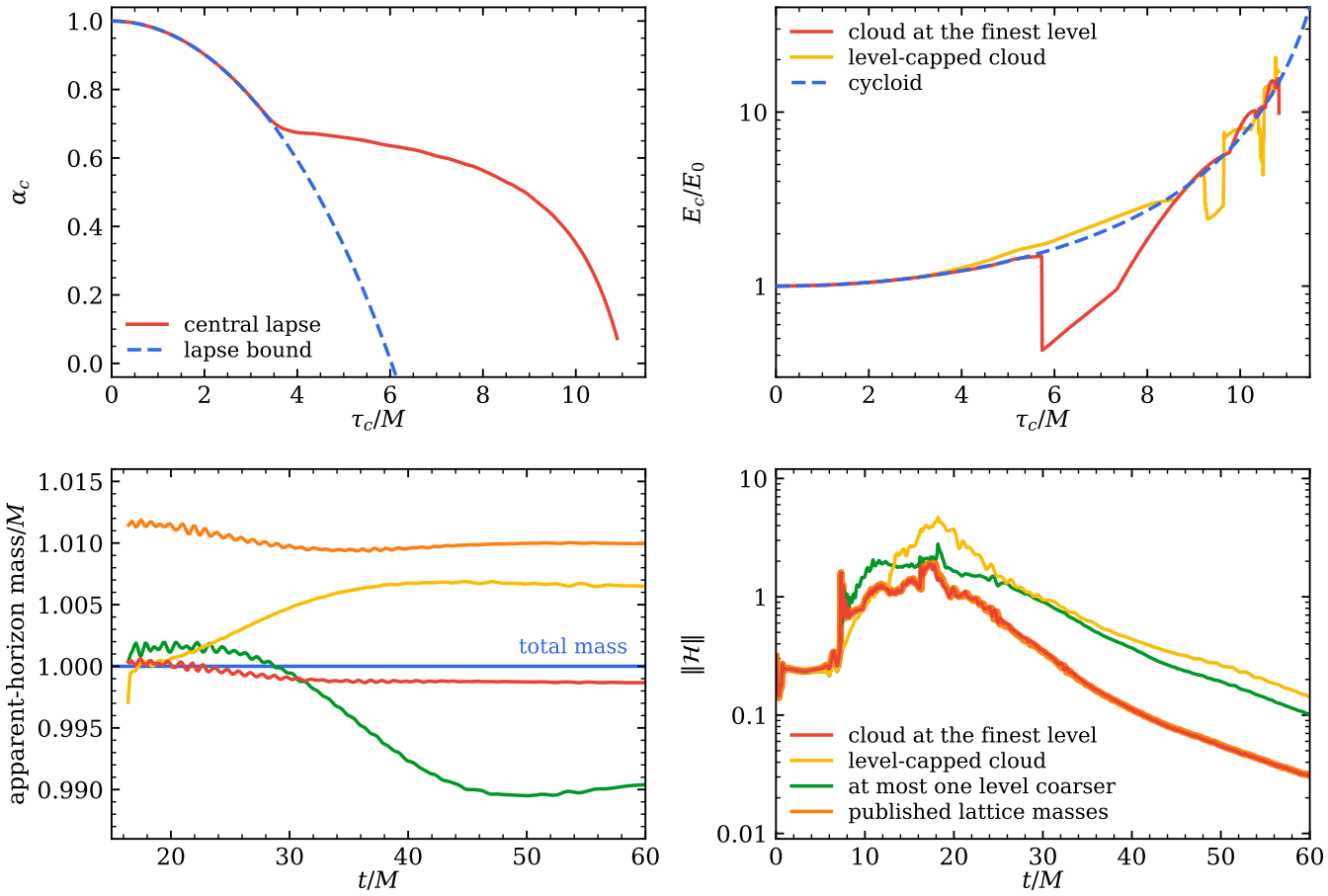


FIG. 8. Gravitational collapse of a pressureless dust ball ($R_0 = 5\,M$, nearly 10^4 macroparticles) to a black hole on an adaptive mesh. Top left: central lapse α_c versus central proper time τ_c (red) and the analytic lapse bound (blue dashed). Top right: central energy density E_c/E_0 and the cycloid (blue dashed) before central excision. Bottom: apparent-horizon mass with total mass M (blue), and Hamiltonian constraint norm $\|\mathcal{H}\|$. Curves show finest-level clouds with surface-corrected (red) or published (orange, bottom) lattice masses, and level-capped clouds with unlimited (yellow) or one-level (green, bottom) coarsening.

low M. A first reading, that the flow iteration stopped too early, was ruled out on an evolved vacuum puncture of mass M (Sec. IV B) when a much tighter stopping rule left the deficit unchanged. Across three cell spacings on that puncture, AthenaK's distributed finder reported masses well below M, with a deficit that is nonmonotonic in cell size (Fig. 9, top panel, solid curves; Table III). On the same slices, we evaluated the slice Misner–Sharp mass $M_{\rm MS}$, the enclosed mass from slice geometry alone (Sec. III F; Fig. 9, open circles). Whereas $M_{\rm MS}$ fell short of M on the coarsest mesh, it matched M on the two finer meshes where the black hole is resolved (Table III). On those two resolved meshes, the evolved spacetime carried the mass M and the deficit belonged to the finder. Inspection of the finder showed that it interpolates metric derivatives stored on active cells, whose ghost zones remain unfilled across block faces (Secs. III F and VI). Relative to $M_{\rm MS}$, the finder deficit is largest on the finest mesh, where blocks are narrowest and more block faces cut the origin-centered horizon. Direct differentiation of the ghost-filled Lagrange metric interpolant on the trial surface eliminates those block-face errors and restores spherical zero-expansion horizons. At the two finer cell sizes, the corrected apparent-horizon mass $M_{\rm AH}$ matches $M_{\rm MS}$ closely on saved slices and in a live run (Fig. 9, dashed curve and diamonds; Table III).

Level-capped deposition (Sec. III C) sets $\ell_{\rm cap}$ from local particle density so sparse refined regions deposit onto virtual cloud cells of width near the initial particle spacing h. On the surface-corrected lattice, an unlimited cap removes the $\|\mathcal{H}\|$ jump when the center refines to $\Delta x = h/4$ (Fig. 8, bottom right; Table IV). Although innermost particle radii stay identical across the runs, the central-density reading $E_c(\tau_c)$ then sits above the cycloid (Fig. 8, top right, yellow curve). That excess arises because the contracting particle lattice beats against virtual cloud cells of nearly equal spacing h at the center. Stored particles from the finest-level run reproduce these E_c readings when redeposited onto width-h clouds. Near the forming black hole, the unlimited cap raises $\|\mathcal{H}\|$ and

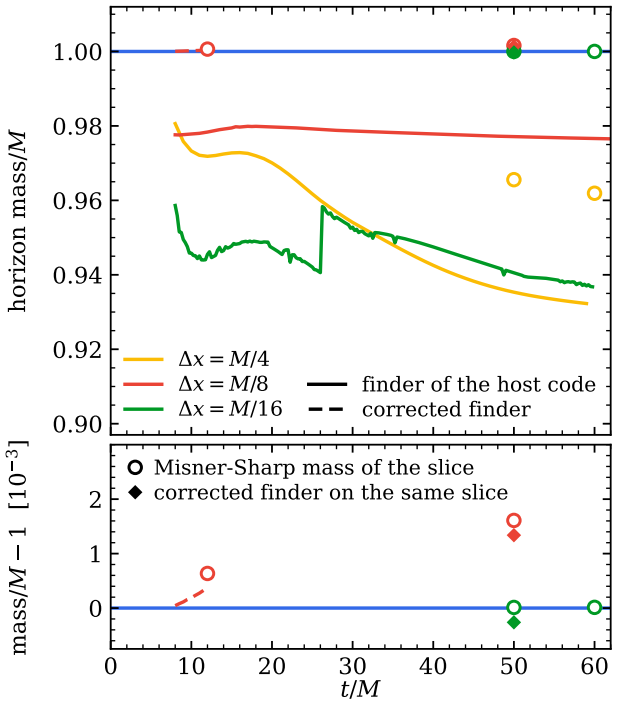


FIG. 9. Apparent-horizon mass of a vacuum black hole of mass M (blue) at three cell spacings (colors). Solid curves show the host code's finder, and the dashed curve differentiates the metric interpolant. Circles show the slice Misner–Sharp mass and diamonds the corrected finder on the same slices (enlarged below).

drives M_{AH} above M after excision. A one-level coarsening cap raises E_c further (Table IV), elevates $\|\mathcal{H}\|$, and shifts M_{AH} to about one percent below M (Fig. 8, bottom panels, green curves).

Randomly sampled dust controls test finest-level against level-capped clouds at two refinement depths (Tables I and IV). At full depth ($\Delta x = h/8$) under plain or compensated filtering, finest-level clouds follow sampling noise past the block limit before an apparent horizon forms. At that depth under either filter, level-capped clouds keep refinement bounded through black hole formation with M_{AH} within a few tenths of a percent of M. One refinement level shallower ($\Delta x = h/4$) with compensated filtering, both finest-level and level-capped runs complete on the same random sample. In that paired comparison without a lattice, level-capped clouds still raise $\|\mathcal{H}\|$ after black hole formation and shift M_{AH} lower by a few tenths of a percent (Table IV). That paired run refuted an initial explanation attributing the coarser-cloud disturbance to the lattice beat alone. Coarser clouds raise constraint residuals near a forming black hole and shift M_{AH} by a few tenths of a percent, more on a lattice, yet bound deep refinement on random samples. Finest-level clouds are more accurate where refinement stays bounded and are used for every production collapse run, while the level-capped runs serve as controls.

B. Prolate collapse

Cold prolate collapse from rest tests the coupled solver in three dimensions beyond spherical symmetry. Transverse infall focuses collisionless matter onto the symmetry axis into a caustic before any apparent horizon forms. Early axisymmetric simulations found rapidly growing curvature ahead of any apparent horizon [54]. Analytic examples showed that spatial slices of a black hole spacetime can miss trapped surfaces [55]. Three-dimensional particle runs later found resolution-dependent curvature peaks inside the spindle for this model [56]. Generalized-harmonic evolutions then followed collapse past the caustic to black hole formation [19]. Particle-in-cell Z4c runs likewise formed an apparent horizon after the caustic [21]. A homogeneous spheroid of mass M and eccentricity $e = 0.9$ starts from rest with $N = 10^4$ symmetric equal-mass macroparticles (Appendix B) on an adaptive mesh (Table I).

Radial infall contracts the spheroid into a thin spindle (Fig. 10, top row, at saved snapshots nearest the published times [21]). At $N = 10^4$ this spindle is an axial chain of separate condensations (Fig. 10, third panel). Equatorial reflection symmetry of the sample places two mirror-image maxima of the Kretschmann scalar I (Sec. III F) inside the matter on the axis, first near the ends [21]. Those maxima move inward in steps as peak I jumps between condensations (Fig. 10, bottom left). Peak I depends on cell spacing [21], with an upward jump in Fig. 10 when the finest mesh levels appear. With grid curvature finite throughout focusing, the run continues through the stage of largest curvature. An apparent horizon first forms at the time shown in Ref. [21], with spindle tips extending past its poles (Fig. 10). Lapse excision below α_{exc} then removes all 10^4 macroparticles as those tips fall into the black hole.

The apparent-horizon mass M_{AH} (Sec. III F) peaks about five percent below M after polar infall (Fig. 10, bottom right; Table V). That apparent-horizon mass continues to fall at the end of the run. Reference [21] states no apparent-horizon mass for this test. The slice Misner–Sharp mass M_{MS} (Sec. III F) on the final slice agrees with the finder at the apparent horizon within its angular spread (Table V). The same profile M_{MS} recovers the total mass M at large radius. The remaining five to eight percent of M after lapse excision reside in gravitational fields outside the black hole (Table V). Outside the black hole, $M_{\mathrm{MS}}(R) - M_{\mathrm{AH}}$ sums radiated gravitational-wave energy and geometric mass from constraint violations without separating them. The Hamiltonian constraint norm $\|\mathcal{H}\|$ peaks at largest curvature about three times above its lattice dust peak (Sec. V A; Table V). A spherical control with a random sample and the same filter loses only a few tenths of a percent of M, so random sampling alone does not explain this deficit. How the deficit of the

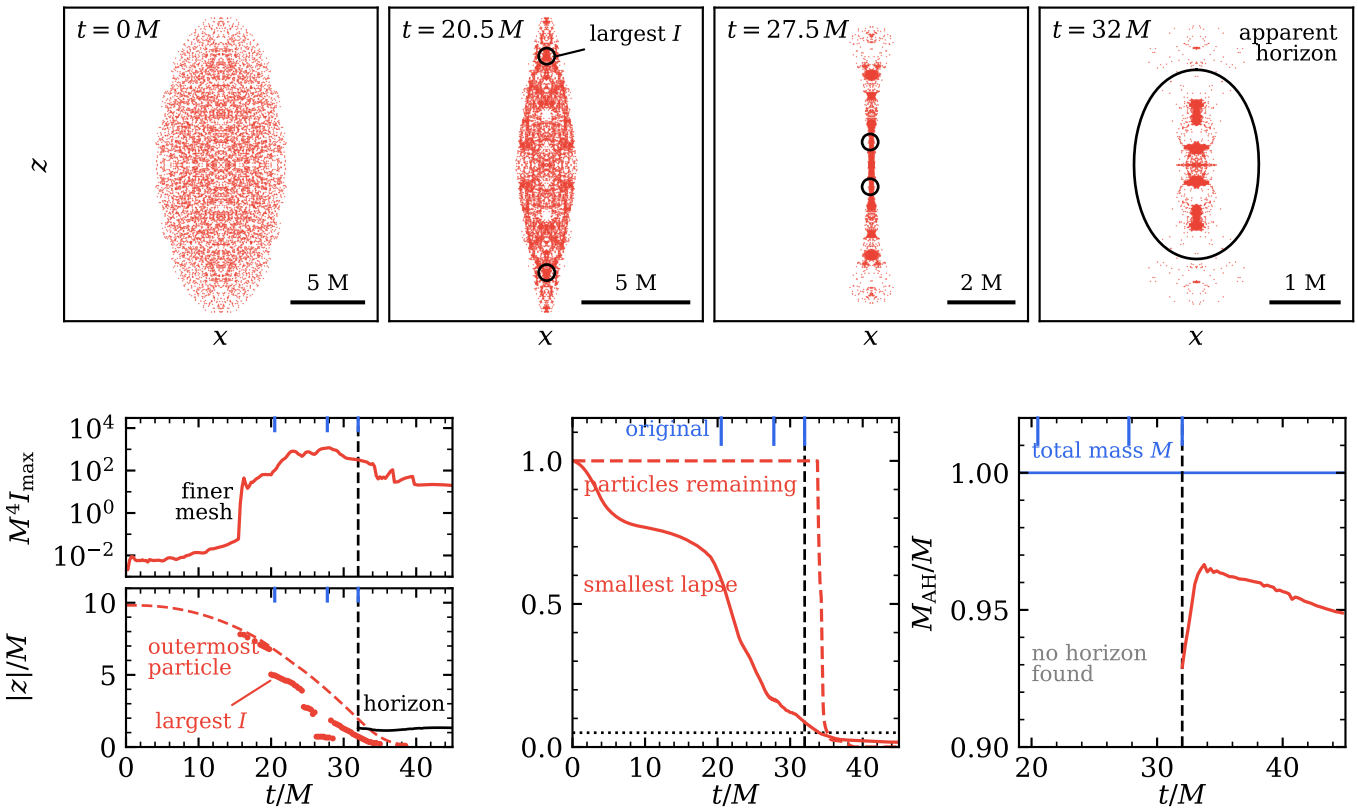


FIG. 10. The cold prolate spheroid ($e = 0.9$, $N = 10^4$) collapses into a segmented spindle whose two axial curvature maxima step inward before an apparent horizon forms. Top: x–z macroparticles, maxima of the Kretschmann scalar I (circles), and first apparent horizon (black). Bottom left: peak I, plus axial $|z|$ of that peak (red), outermost macroparticle (red dashed), and horizon (black). Bottom middle and right: smallest lapse with $\alpha_{\rm exc}$ (dotted) and surviving fraction (red dashed), and $M_{\rm AH}$ with M (blue). Vertical dashed lines mark first horizon; blue ticks mark snapshot times [21].

prolate collapse splits between particle number and cell size at its caustic stage is not determined. Dynamic refinement (Sec. III E) with halved finest cells exceeds the block memory of one graphics card, which sets our finest grid.

On the prolate sample at $\Delta x = M/32$, four binomial passes with compensation (Sec. III C) remove only the early noise-driven refinement. Curvature along the collapsing spindle itself still drives the block count past the memory limit before an apparent horizon forms. The production run stays bounded because its refinement depth is limited by one level ($\Delta x = M/16$) alongside that compensated filter (Table I).

C. Relativistic clusters

Warm collisionless star clusters in equilibrium balance collective gravity through velocity dispersion. Their evolution tests phase-space conservation alongside general-relativistic radial stability [1, 10–12]. As in Shapiro and Teukolsky [1, 12] and Ref. [21], static, spherically symmetric, isotropic monoenergetic (single-energy) clusters populate one Killing energy $\mathcal{E} = \alpha W = \alpha_{\rm s}$. Here $\alpha_{\rm s}$ is the lapse at the cluster surface, where particle speed vanishes ($W = 1$). Radial integration of the Einstein-Vlasov equilibrium relations—the structure equations—yields a one-parameter sequence labeled by $y_c \in (0, 1)$, with gravitational binding energy E_b,

$$y_c = \left(\frac{\alpha_{c,\rm eq}}{\alpha_{\rm s}}\right)^2, \qquad E_b = M_0 - M. \tag{43}$$

In this sequence, $\alpha_{c,\rm eq}$ is the equilibrium central lapse, $z_c = \alpha_{c,\rm eq}^{-1} - 1$ is the central redshift, M_0 is total rest mass, and M is ADM mass [1, 3]. From diffuse, nearly Newtonian clusters ($y_c \to 1$, low z_c) to compact, relativistic cores (smaller y_c, high z_c), E_b/M_0 rises to one maximum along the sequence. At still smaller y_c, E_b decreases and turns negative ($M > M_0$). Like the turning point of fluid stars [57, 58], that binding-energy maximum separates stable clusters at larger y_c from unstable clusters at smaller y_c [11, 12, 21]. Perturbed clusters on the high-y_c branch hold together, whereas clusters past the maximum collapse or expand. Each unit-mass cluster draws up to 10^4 equal-mass macroparticles with zero net momentum (Sec. III F). Static meshes evolve the two

loosest models and adaptive meshes evolve the kicked compact ones, with all meshes and particle numbers in Table I.

Multi-pass grid filtering suppresses discrete source fluctuations at $N = 10^4$, yet uncompensated smoothing biases self-gravitating equilibria outward. An unperturbed cluster on the stable branch ($y_c = 0.819$) was expected to stay near equilibrium with minor oscillations, as for the kicked model in Ref. [21]. With four plain binomial passes, the cluster instead expanded by a visible fraction of its radius and dropped in central density before rebounding. Over that bounce, α_c rose well above $\alpha_{c,\mathrm{eq}}$ (Fig. 11, left panel, solid curves; Table VI). A first reading, that discrete sampling noise drove this excursion, was ruled out when three independent 10^4-particle samples traced the identical bounce up to a tiny ripple (Fig. 11, left panel). Sweeps across zero, two, and four plain passes on one sample showed that the initial lapse rise grows systematically with pass count n_f (Fig. 11, middle panel). Each plain binomial pass widens every cloud's second moment, which acts as a gravitational softening over the filter scale [2] (Sec. III C). Plain passes also lower the proper-volume integral of E slightly across nonuniform $\sqrt{\gamma}$. Because particle velocities match the unsoftened equilibrium, that shallower smoothed potential lets the cluster expand and recontract. The compensation pass of Sec. III C [6] cancels that second-moment softening and preserves the proper-volume energy integral. It removes the bounce across two and four passes and leaves $\alpha_c(t)$ insensitive to n_f (Fig. 11, dashed curves). Unfiltered evolution ($n_\mathrm{f} = 0$) also avoids the bounce, yet leaves $\|\mathcal{H}\|$ much higher (Table VI) and causes α_c to wander downward later (Fig. 11, middle panel). All cluster runs below therefore combine four binomial passes with compensation.

Dynamically stable clusters on the high-y_c side of the binding-energy maximum are expected to remain bound and oscillate stably under small radial perturbations. As in Ref. [21], we apply a one-percent inward velocity kick to the stable models $y_c = 0.900$ and $y_c = 0.819$ at $N = 10^4$ (Fig. 12, first two columns). Over several crossing times, both clusters retain every macroparticle and stay centered near the origin, as in Ref. [21]. Rest-mass radii stay within a narrow band for $y_c = 0.900$ and undergo bounded radial swings in the inner shells of $y_c = 0.819$ (Fig. 12, top left). Because its potential is shallower, the looser model $y_c = 0.900$ shows much smaller variations in α_c (Table VI) and mean Lorentz factor than $y_c = 0.819$.

Closer inspection of the stable model $y_c = 0.819$ reveals a coherent radial oscillation of its core that grows before saturating at late times. With filter softening removed, an unperturbed equilibrium was expected to hold a steady central lapse $\alpha_c = \alpha_{c,\mathrm{eq}}$. Instead, α_c oscillates at a period close to the radial period of a particle in the core (Fig. 13, top panel; Table VI), while the outer shells stay within a few percent of their initial radii over the first three hundred M and move more afterward. Because a nearly uniform core creates an almost harmonic potential with a shared radial period, collective core motion resists phase mixing [2]. Over the first several periods the swing grows steadily, so a long continuation tested unbounded instability against saturation. In that extended run, growth halts after several cycles and the swing stays bounded at a small fraction of $\alpha_{c,\mathrm{eq}}$ with all 10^4 particles retained (Fig. 13, top panel). A first reading, that intermediate-wavenumber sampling noise passed by the compensated filter excited this mode, was ruled out by three stratified samples (Fig. 13, top panel) and a Monte Carlo sample. All samples share the same period, phase, and growth. Inward-kicked runs at $N = 10^3$ and 10^4 likewise show matching early swings without $N^{-1/2}$ amplification (Fig. 13, bottom left; Table VI).

Further controls on the first sample tested whether time-stepping lag, constraint handling, cloud width, or cell spacing drives this core oscillation (Fig. 13, bottom middle and right; Table I). Frozen sources across Δt could lag particle motion, yet a halved step $\Delta t/2$ leaves $\alpha_c(t)$ unchanged. Feedback from constraint damping or advected constraint residuals was tested across raised and zero κ_1 and with parabolic relaxation of the Hamiltonian constraint (Sec. III D). Although those settings alter $\|\mathcal{H}\|$ substantially, all three reproduce $\alpha_c(t)$ to a tiny fraction of the swing (Fig. 13, bottom right; Table VI). An extended compensation weight c that also cancels cloud width trims the slow excursion slightly while leaving the oscillation intact (Fig. 13, bottom middle; Table VI). A control with halved cell spacing over the cluster (Table I) leaves the first α_c maximum unchanged, gives an equal or slightly larger growth rate, and drifts further in mean lapse. That control refutes a second-order cell-size origin for the oscillation. In the sequence figure of Ref. [21], the full-range lapse axis is too coarse to resolve a swing of 0.01 in $y_c = 0.819$. By contrast, the narrower-axis figure for the perturbed near-critical cluster in Ref. [21] at $N = 10^6$ displays the same growth, saturation, and decay at the core period. What sets the small common initial offset from which the unperturbed oscillation starts here is not determined.

Smaller counts N at fixed mass M amplify force fluctuations as $N^{-1/2}$ and hasten collisional heating. Stratified and Monte Carlo runs of $y_c = 0.819$ across $N = 10^4$, 10^3, and 10^2 locate where graininess overtakes collective dynamics (Table I). At $N = 10^3$, the cluster holds together and follows the $N = 10^4$ oscillation across several periods (Fig. 13, bottom left). Later cycles at $N = 10^3$ develop deeper lapse minima and a slow downward drift. At $N = 10^2$, discrete fluctuations overtake the cluster within one period. Net momentum wanders, mean kinetic energy more than doubles, and α_c plunges steadily as equilibrium breaks down (Table VI). Both the $N = 10^4$ and 10^3 clusters hold across the comparison run, whereas the $N = 10^2$ cluster loses equilibrium.

Past the binding-energy maximum the clusters are unstable [11, 12]. Reference [21] finds collapse under an inward kick for the bound models past that maximum ($E_b > 0$) and dispersal for the unbound model

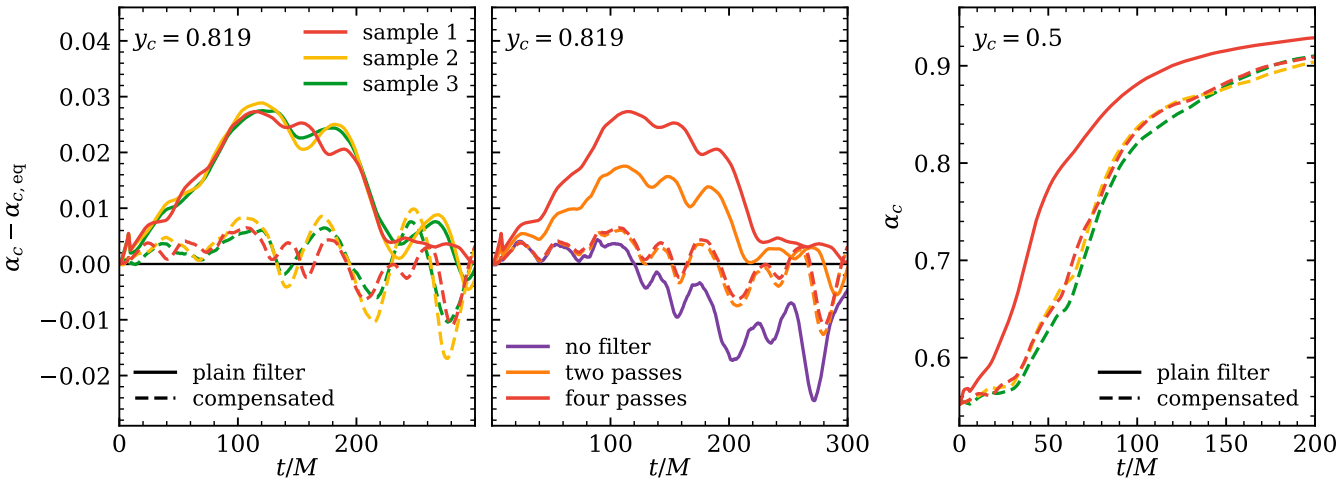


FIG. 11. Source filtering and radial motion of monoenergetic clusters with $N = 10^4$ macroparticles. Left: central-lapse offset $\alpha_c - \alpha_{c,\mathrm{eq}}$ from equilibrium (black line) for the stable model $y_c = 0.819$ across three samples (colors), comparing four plain binomial passes (solid) with compensated filtering (dashed). Middle: one sample with zero, two, and four filter passes (colors), plain (solid) and compensated (dashed). Right: central lapse α_c of the unperturbed unstable model $y_c = 0.5$, encoded as on the left.

($E_b < 0$). With the one-percent inward kick on adaptive meshes, both bound unstable models $y_c = 0.605$ and $y_c = 0.500$ collapse to black holes (Fig. 12, third and fourth columns). Once infall accelerates, mass shells contract together, α_c plunges toward zero, and an apparent horizon forms. As outer shells accrete, M_{AH} rises rapidly to M while lapse excision removes almost every macroparticle (Fig. 12, bottom panels; Table VI). Black hole formation and full mass capture agree with Ref. [21], alongside two documented differences. First, apparent horizons form later here for both $y_c = 0.605$ and $y_c = 0.500$ than in Ref. [21] (Table VI). Reference [21] notes that these collapse times could depend on N and initial sampling noise. In $y_c = 0.605$, a growing core oscillation also precedes collapse (Fig. 12, third column), whereas the published lapse falls smoothly. The cause of these later collapse times is not determined. Second, after horizon formation α_c settles closer to zero here than in Ref. [21]. This difference concerns the coordinates at the center, and its origin was not looked into. For the unbound cluster $y_c = 0.283$ ($E_b < 0$), α_c dips briefly after the kick before rebounding into free dispersal (Fig. 12, rightmost column). Its central lapse approaches unity and mass shells expand at constant asymptotic speeds with every macroparticle kept. This free dispersal agrees with Ref. [21], whereas early spherical runs [12] reported black hole formation.

Without a kick, an unstable cluster past the binding-energy maximum can expand dynamically toward the stable branch of equal mass, a nonlinear transition termed migration [21, 39]. Because the same model $y_c = 0.5$ expands when unperturbed in Ref. [21] whereas kicked $y_c = 0.500$ collapses, we first asked whether plain-filter softening alone selected expansion over collapse. Paired runs of unperturbed $y_c = 0.5$ with plain and compensated four-pass filtering tested that hypothesis directly (Fig. 11, right panel). Plain filtering advances the expansion noticeably, yet three compensated samples still expand outward on a common schedule (Fig. 11, right panel). Plain filtering therefore hastens an already outward transition. In the full run on a static mesh with compensation, the unperturbed cluster $y_c = 0.5$ expands strongly and then recontracts deeply (Fig. 14; Table I). After that rebound, the cluster settles into a damped oscillation about a more extended state to the end of the run. The sequence of states and the first two maxima and minima of α_c agree with the migration curve of Ref. [21] (Table VII). After the first maximum, every extremum of α_c arrives earlier here than in Ref. [21], and the later swings are smaller. Once the cluster expands, most matter lies outside the finest central box of this static mesh on coarser cells, whereas Ref. [21] refines adaptively (Table I). Whether the coarser outer mesh, lower N, or filter compensation causes those timing and amplitude differences is not determined. Over the run, four macroparticles leave the domain and none is removed by lapse excision (Table VII).

Near the binding-energy maximum, Ref. [21] applies a ten-percent initial density perturbation to $y_c = 0.756$ at $N = 10^6$. It reports reconfiguration into a persistent core–halo structure with all macroparticles retained, damped early lapse oscillations, and a slow fall of α_c. Its text states the amplitude of the perturbation and leaves its sign and the initial-metric construction open. Over the shorter runs here, the comparison is limited to α_c and the enclosed rest-mass radii shown in Fig. 15. Here homologous radial scaling changes the density by ten percent of either sign while ψ solves $\mathcal{H} = 0$ for the perturbed

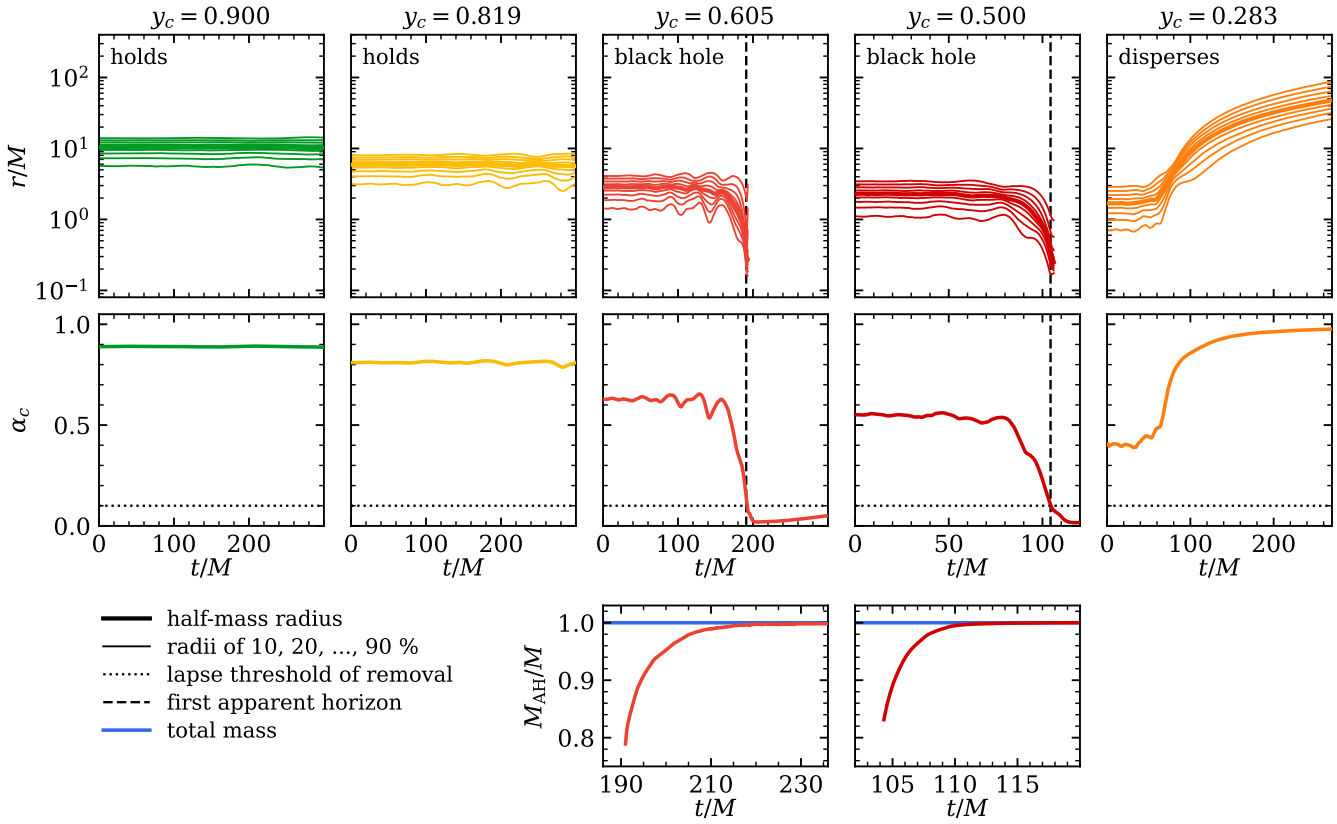


FIG. 12. Five monoenergetic clusters ($N = 10^4$) with an inward velocity kick, ordered from least compact (left) to most compact (right). Models $y_c = 0.900$ and 0.819 hold, 0.605 and 0.500 form black holes, and the unbound model 0.283 disperses. Top: coordinate radii enclosing tenths of remaining rest mass (half-mass thicker) while at least half remains. Middle: central lapse α_c with excision threshold $\alpha_{\rm exc}$ (dotted) and first apparent horizon (dashed, also top). Bottom: apparent-horizon mass $M_{\rm AH}$ and total mass M (blue).

matter (Appendix B). Under ten-percent compression, the cluster contracts and collapses to a black hole with every macroparticle removed by lapse excision (Fig. 15; Tables I and VII). The final apparent-horizon mass equals the ADM mass of the compressed initial data. A control run with the compressed particles on the unperturbed equilibrium metric contracts in the same way. Under ten-percent expansion, the cluster oscillates and keeps every macroparticle (Fig. 15). Its oscillation runs about a higher central lapse and with a slower rhythm than the published α_c curve of Ref. [21]. The unperturbed model $y_c = 0.756$ oscillates about its equilibrium at the core period with a swing that grows and then stops growing (Fig. 15). Its mean lapse falls slowly below the equilibrium value over the run, as in the published α_c curve. That α_c evolution matches the core oscillation of the stable model in Fig. 13 and, apart from a time shift, resembles the published α_c curve in Ref. [21] most closely (Table VII). As an interpretation, the oscillation in the published figure may be this intrinsic core oscillation, while which perturbation Ref. [21] applied cannot be decided from its text.

VI. HOW THE CODE WAS ASSEMBLED

The published text of the numerical-relativity particle-in-cell (NRPIC) method [21] provided the sole specification for the coupled Einstein-Vlasov solver built here without access to the original authors' code. This independent reconstruction tests whether the written equations and algorithmic rules fix the coupled method completely. The implementation assembles the coupled solver inside the host code AthenaK [35, 36] from existing capabilities across neighboring fields. AthenaK supplies the block-structured octree mesh with adaptive refinement, ghost-zone exchange, prolongation, and restriction. It also provides the Z4c spacetime solver in moving-puncture gauge, the particle container and drift update, the magnetohydrodynamics matter-source interface, and the pseudospectral apparent-horizon finder [35, 36]. Portable parallel execution of mesh stencils and particle loops comes from the Kokkos performance-portability library [37].

Plasma particle-in-cell methods contribute implicit midpoint geodesic integration [27], the velocity-space

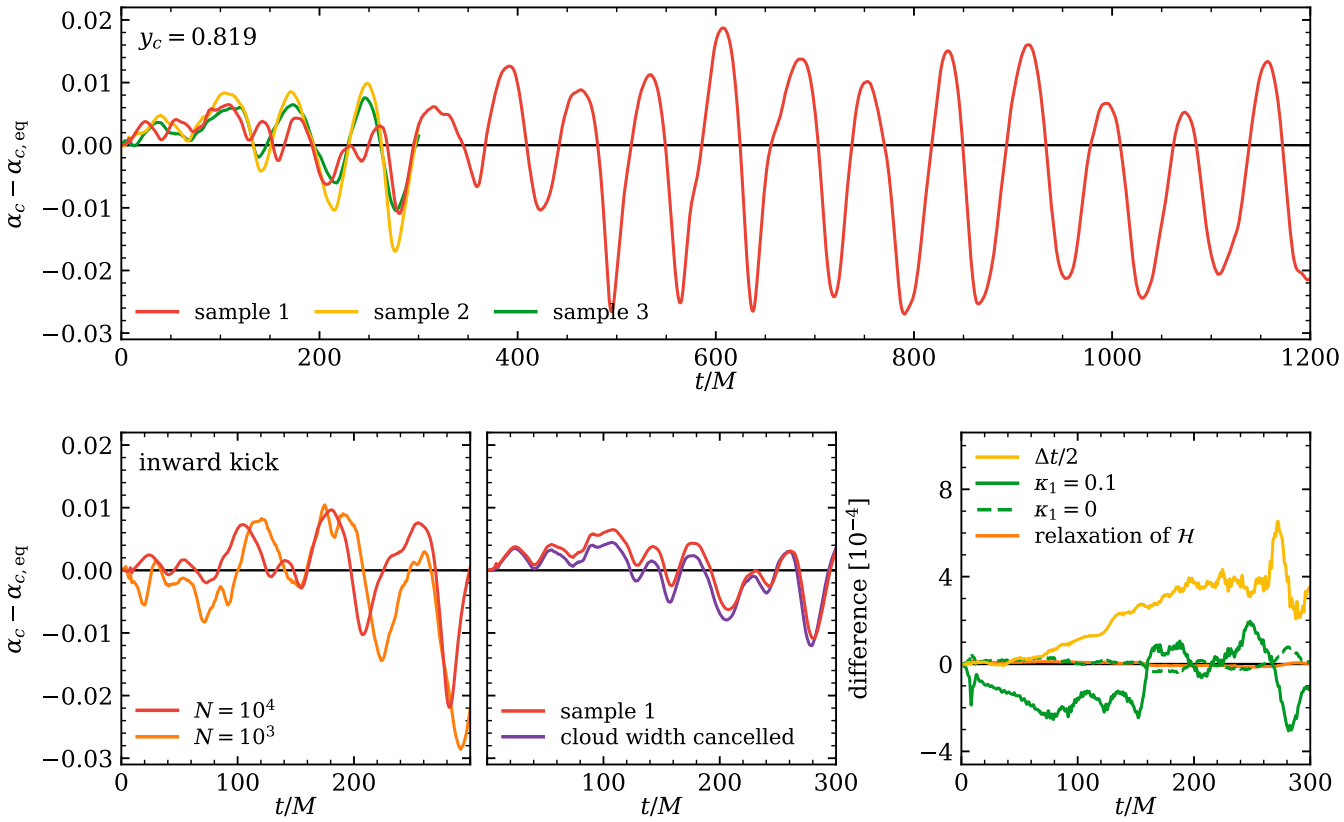


FIG. 13. Core oscillation of the stable cluster $y_c = 0.819$ in central-lapse offset $\alpha_c - \alpha_{c,\mathrm{eq}}$ from equilibrium (black lines). Top: three unperturbed samples at $N = 10^4$ (colors), with sample 1 continued to saturation. Bottom left: inward-kicked runs at $N = 10^4$ and 10^3. Bottom middle: sample 1 against compensation that also cancels cloud width. Bottom right: differences from sample 1 for $\Delta t/2$, raised and zero (dashed) constraint damping κ_1, and relaxation of the Hamiltonian constraint.

Boris rotation [6, 26], cloud-in-cell deposition with binomial filtering and compensation [6, 22], and conservative single-level deposition across refinement boundaries [30]. Numerical relativity and mesh methods supply the tetrad form of geodesic motion [28, 29], lapse excision inside black holes [49], parabolic relaxation of the Hamiltonian constraint [45, 46], and the Löhner refinement indicator [48]. Exact analytical solutions for Kerr geodesics, Oppenheimer–Snyder dust collapse [38], monoenergetic cluster equilibria [1], and homogeneous spheroid potentials complete the external inputs. This work combines those inputs into the two geodesic pushers in the symmetric and Cholesky tetrad gauges, cubic Hermite metric reconstruction, and single-level deposition with an optional level cap and compensated filtering. Additional components built here provide particle-coupled adaptive mesh refinement, lapse excision with interpolant-derivative apparent-horizon diagnostics, constraint-control terms, initial-data and validation suites, and the staggered coupled cycle.

Figure 16 shows which capabilities each component combines. Horizontal rows list the existing capabilities grouped along the left margin by their origin. Vertical red columns represent the components built in this work as hyperedges of a hypergraph. Blue dots along each column mark the inherited capabilities joined by that hyperedge. Red rows at the bottom denote the geodesic-pusher and deposition components built here. Red dots on those bottom rows mark their reuse inside the adaptive-mesh and coupled-cycle hyperedges.

Large-language-model agents carried out the reconstruction under human direction across dedicated solver-development, verification, and manuscript roles. One model in the solver-development role wrote the derivations and solver source and directed the tests. Other models in the verification role built, ran, and analyzed the simulations, wrote independent reference solutions without access to the solver source, and audited the derivations, figures, and tables. A third model in the manuscript role wrote the text from the method specifications, figures, and test outcomes.

Candidate explanations for unexpected simulation results were tested against predictions recorded in advance. Each control run designed to test a stated explanation was preceded by a written prediction of the expected outcome. Seven initial explanations were refuted by these control runs. First, an apparent-horizon mass below M after spherical collapse was attributed to early stopping

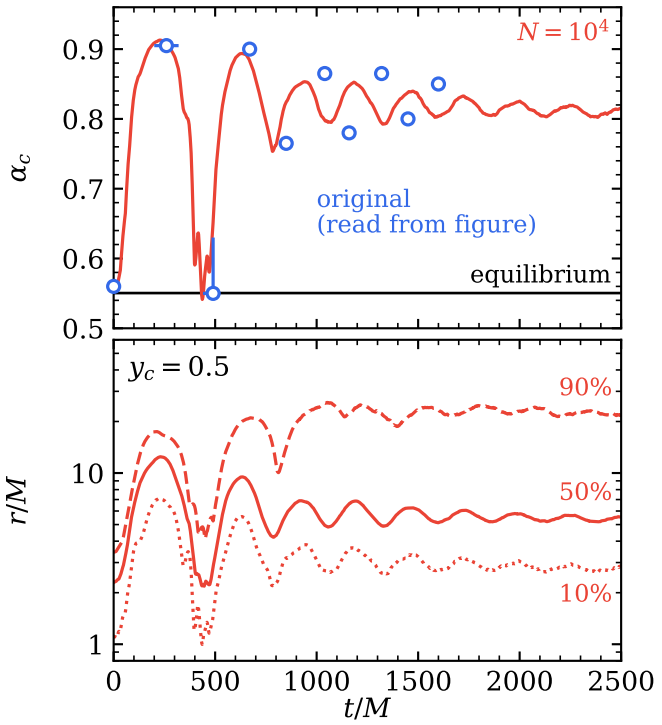


FIG. 14. The unperturbed cluster $y_c = 0.5$ expands, recontracts deeply, and oscillates with decreasing swing about a more extended state. Top: central lapse α_c, equilibrium (black), and extrema from Ref. [21] (circles; bars: read ranges). Bottom: radii enclosing the labeled rest-mass fractions.

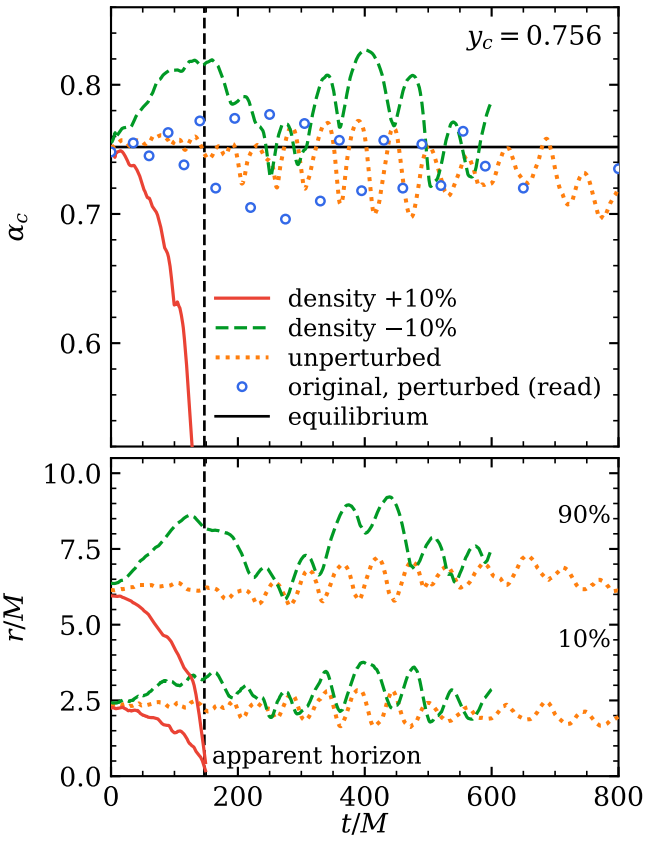


FIG. 15. The near-critical cluster $y_c = 0.756$ collapses to a black hole when compressed, oscillates when expanded, and shows the core oscillation when unperturbed. Top: central lapse α_c, first apparent horizon (dashed), and extrema from Ref. [21] (circles). Bottom: radii enclosing the labeled rest-mass fractions.

of the pseudospectral flow finder. Calibration runs on an evolved vacuum black hole of known mass M tested that explanation with a tighter stopping rule, which left the deficit unchanged. On the two finer meshes where the black hole is resolved, independent evaluations of the slice Misner–Sharp mass M_{MS} confirmed that the slice carried the mass M (Fig. 9; Table III). Direct inspection of the host code's finder showed that it interpolates a cell-centered metric-derivative array whose ghost zones retain their allocated zeros across mesh-block faces. Differentiation of the ghost-filled metric interpolant at those two cell sizes brought M_{AH} into agreement with M_{MS} on saved slices and in a live run. In Sec. V A, three further initial explanations failed predictive controls: collapse dynamics for post-excision mass drift, harmlessness of one-level cloud coarsening, and the lattice beat alone for coarser-cloud disturbances. In Sec. V C, three more failed: sampling noise for the plain-filter bounce, sampling noise for the core oscillation, and second-order cell-size error for that oscillation. Independent audits of initial-data code and appendix tables against saved outputs caught two derivation slips and exposed unsupported log entries that were then corrected. In particular, redeposition from stored particles showed that a central-density excess under coarser clouds, first logged as matching the cycloid, came from the particle lattice beating against virtual cloud cells. Every effect without an isolating run has its cause stated in Sec. V as not determined.

All materials needed to inspect and extend the simulations are public alongside the solver source code. The public release includes the simulation inputs, the analysis scripts, and the chronological record of every prediction and outcome. Readers can rerun the validation benchmarks, modify individual stages of the particle-mesh cycle, or audit each diagnostic test from those materials. The data-and-code-availability statement before the appendices gives the archive location.

DATA AND CODE AVAILABILITY

The solver source (a particle module inside the public code AthenaK [35, 36]), the inputs of every run, and the initial-data generators are public at `https://github.com/FoAKTEE/nrpic`. The same repository provides the

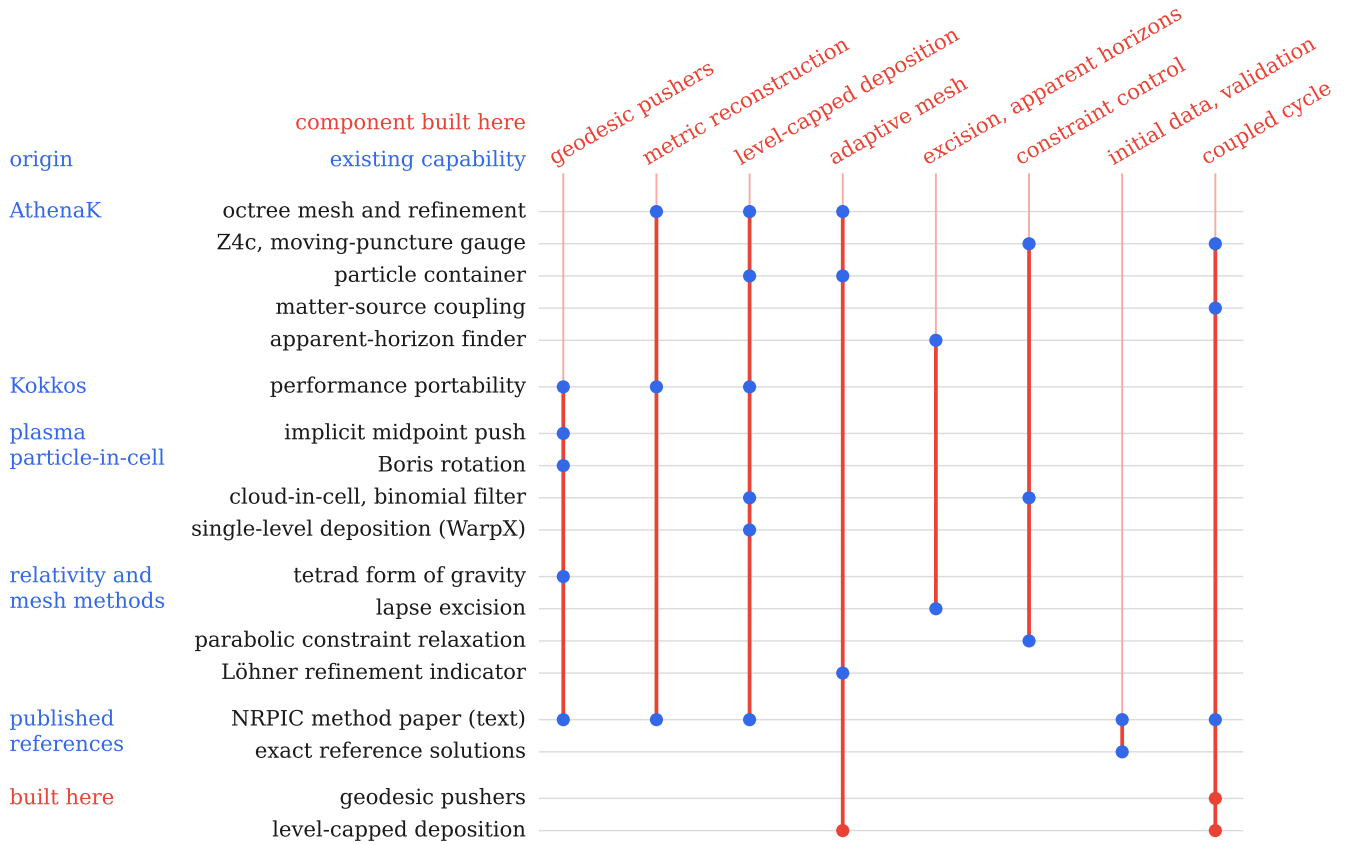


FIG. 16. The hypergraph shows how the code was assembled. Each row is a capability taken from an existing code or published method, grouped by origin. Each red column is a component built here, a hyperedge joining the capabilities it combines (dots). The red rows at the bottom are built components reused inside the coupled cycle and the adaptive mesh.

analysis and figure scripts alongside the data behind every figure. It also holds the ledger of every number in the tables and the chronological record of predictions and outcomes.

ACKNOWLEDGMENTS

This work rebuilds the method of Ref. [21] from its published text, and we thank its authors for the level of detail of that text. We acknowledge the developers of AthenaK [35, 36] and Kokkos [37] for the public host code and performance-portability library used here. Large-language-model agents under human direction wrote the solver source, ran and analyzed the simulations, made the figures, and wrote the text. One language-model family served each of the dedicated solver-development, verification, and manuscript roles described in Sec. VI. The human authors directed the work and decided what was tested. They take responsibility for the content of this paper.

[Funding and computing-resource acknowledgments to be supplied by the authors.]

Appendix A: Scenario parameters and measured values

Table I lists the setup of every family of runs, and Table II lists the refinement, horizon-finder, and pusher settings. Tables III–VII list the measured values to which the text refers, with the values of Ref. [21] where it gives them.

Appendix B: Initial data and sampling

Coupled Einstein-Vlasov evolutions require constraint-satisfying initial slices alongside discrete particle ensembles that sample the continuum phase-space distribution. Each self-gravitating run starts on a conformally flat, time-symmetric slice where superposed flat-space Poisson potentials u_c for conformal sources $\tilde{\rho}_c$ solve $\mathcal{H} = 0$ via Eq. (36). Around each component center $\boldsymbol{x}_c$ at isotropic distance $\bar{r}_c = |\boldsymbol{x} - \boldsymbol{x}_c|$, four component families enter Eq. (36). Tabulated spherical profiles interpolate $u_c(\bar{r}_c) = \psi_c(\bar{r}_c) - 1$ on a uniform grid with four-point Lagrange polynomials and follow Eq. (37) outside with the component ADM mass. Vacuum black hole punc-

TABLE I. Setups, one row per family of runs (numerical settings in Table II): macroparticle number N and sampling, domain half-width L, mesh and cell spacing Δx in the matter at the start and finest reached (vacuum: at the puncture), first step Δt, filter passes n_f (+c: compensation), deposition, excision threshold α_exc, constraint damping κ_1, final time. Non-uniform meshes: 64^3 root cells, 16^3-cell MeshBlocks. Lists follow the model column; parentheses: control runs.

Scenario and model	N, sampling	L/M	Mesh, $\Delta x/M$	$\Delta t/M$	n_f	Deposition	α_exc	κ_1	t_end/M
Kerr geodesics: $a = 0.9M$, $a \to 0$; bound, $e = 0.99$, plunging	1	10–12	uniform, 1/8–1/2; lattice 0.025	0.00125–0.08	–	–	–	–	77.5–4550
Test-particle clusters: $a = 0.9M$, $a \to 0$; Gaussian cloud about $r_0 = 10M$, widths $0.5M$, 0.02 in velocity	10^2–10^4, Gaussian	64	static, 1/4→1/16	0.05	2	finest level	0.62	–	500–1000
Vacuum black hole: puncture of mass M; 6 tracers	6, tracers	128	static, 1/4, 1/8, 1/16	1/16, 1/32, 1/64	–	none	0.05	0.02	12–250
Head-on collision: punctures $M/2$ at $x = \pm 5M$; 6 tracers	6, tracers	128	adaptive, 1/16	1/64	–	none	0.05	0.02	160
Dust collapse, lattice: $R_0 = 5M$; $n_\mathrm{lat} = 26$ ($h = 0.30M$); masses corrected, published	9328, lattice	155	adaptive, 0.30→0.038	0.076	1	finest level	0.1	0.02	60
Dust collapse, lattice, level cap: as above, corrected masses	9328, lattice	155	adaptive, 0.30→0.038	0.076	1	level-capped (≤ 1, ≤ 2, any)	0.1	0.02	60
Dust collapse, random sample, full depth: $R_0 = 5M$; positions drawn with density $\propto \psi^6$, equal masses[a]	9328, Monte Carlo	155	adaptive, 0.30→0.038	0.076	1, 4+c	level-capped	0.1	0.02	41–43[b]
Dust collapse, random sample, one level fewer: as above	9328, Monte Carlo	155	adaptive, 0.30→0.076	0.076	4+c	finest level, level-capped	0.1	0.02	60
Prolate collapse: spheroid $e = 0.9$, $b = 10M$, at rest	10^4, Monte Carlo	128	adaptive, 1/4→1/16	1/16	4+c	finest level	0.05	0.02	45
Stable cluster, filter: $y_c = 0.819$, unperturbed; seeds 1–3	10^4, stratified	1024	static, 1/2	1/8	0, 2, 4, 2+c, 4+c	finest level	0.1	0.02	300
Stable cluster, oscillation: $y_c = 0.819$, unperturbed; seed 1	10^4, stratified	1024	static, 1/2 (1/4–1/2)	1/8 (1/16)	4+c[c]	finest level	0.1	0.02 (0.1, 0)[d]	300–1200
Particle number: $y_c = 0.819$; kick 0.01 (stratified) or none (Monte Carlo)	10^2–10^4, stratified, Monte Carlo	1024	static, 1/2	1/8	4+c	finest level	0.1	0.02	300–500
Sequence, stable: $y_c = 0.900, 0.819$; kick 0.01	10^4, stratified	1024	static, 1/2	1/8	4+c	finest level	0.1	0.02	300
Sequence, unstable: $y_c = 0.605, 0.500, 0.283$; kick 0.01	10^4, stratified	512	adaptive, 1/4→1/16	1/16	4+c	finest level	0.1	0.02	120–300
Unperturbed unstable: $y_c = 0.500$, unperturbed; seeds 1–3	10^4, stratified	512	static, 1/4	1/16	4, 4+c	finest level	0.1	0.02	200–2500
Near-critical, consistent data: $y_c = 0.756$; density ×1.1, ×0.9	10^4, stratified	512	adaptive, 1/4→1/16, 1/4	1/16	4+c	finest level	0.1	0.02	160–600
Near-critical, static mesh: $y_c = 0.756$; unperturbed, or ×1.1 on its metric	10^4, stratified	1024	static, 1/4	1/16	4+c	finest level	0.1	0.02	800–818

[a] Finest-level clouds at the full refinement depth passed the MeshBlock limit before a horizon formed: $t = 8.8M$, 1884 asked for, limit 1800 ($n_\mathrm{f} = 1$); $t = 10.2M$, 2024 asked for, limit 2000 ($n_\mathrm{f} = 4 + c$).
[b] Both runs stopped at the time limit of their job, at $t = 43M$ and $t = 41M$.
[c] One control uses $c = n_\mathrm{f}/4 + 1/12$, which also cancels the cloud width.
[d] One control adds parabolic relaxation of $\mathcal{H}$ with constant $c_H = 0.05\,M$ and no Δx-proportional term.

TABLE II. Numerical settings of the families of Table I: refinement and number of levels, regions held at least at the given cell spacing (half-widths; three numbers: x, y, z), MeshBlock limit, apparent-horizon finder, pusher iteration, and the level cap of the capped runs.

Scenario[a]	Refinement[b]	Levels	Minimum resolution: $\Delta x/M$ for $\lvert x\rvert/M$ below	MeshBlock limit	Finder: start t/M (first radius/M)[c]	Pusher tolerance, iterations	Level cap: particles per cloud cell, coarsest cloud $\Delta x/M$
Kerr geodesics	uniform	1	–	–	–	2×10^{-14}, 200	–
Test-particle clusters	static	6	1/4: 24, 24, 8; 1/16: 2	–	–	10^{-12}, 100	–
Vacuum black hole	static	5, 6, 7	1/4, 1/8, 1/16: 1.5	512	0 (0.8)	10^{-10}, 100	–
Head-on collision	puncture tracking	7	1/16: 0.5 about each puncture; 1/2: 15	1024	40 (1)	10^{-10}, 100	–
Dust collapse, lattice	Löhner	8	0.30: 6	640	8 (1)	10^{-7}, 50	–
Dust collapse, lattice, level cap	Löhner	8	0.30: 6	640	8 (1)	10^{-7}, 50	1, 0.30
Dust collapse, random sample, full depth	Löhner	8	0.30: 6	1800, 2000	8 (1)	10^{-7}, 50	1, 0.30
Dust collapse, random sample, one level fewer	Löhner	7	0.30: 6	2100	8 (1)	10^{-7}, 50	–; 1, 0.30
Prolate collapse	Löhner	7	1/4: 5, 5, 11 and $r < 11$	2000	20 (1)	10^{-7}, 50	–
Stable cluster, filter	static	7	1/2: 15.9	1024	–	10^{-7}, 50	–
Stable cluster, oscillation	static	7 (8)	1/2: 15.9 (1/4: 7.9; 1/2: 15.9)	1024	–	10^{-7}, 50	–
Particle number	static	7	1/2: 15.9	1024	–	10^{-7}, 50	–
Sequence, stable	static	7	1/2: 15.9	1024	–	10^{-7}, 50	–
Sequence, unstable	Löhner	9	1/4: 6	1024	10 (1)	10^{-7}, 50	–
Unperturbed unstable	static	7	1/4: 7.9	1024	–	10^{-7}, 50	–
Near-critical, consistent data	Löhner	9	1/4: 6 and $r < 8$	1024	10 (1)	10^{-7}, 50	–
Near-critical, static mesh	static	8	1/4: 7.9	1024	–	10^{-7}, 50	–

[a] Every Z4c run: fourth-order Runge–Kutta steps $\Delta t = 0.25\,\Delta x_{\min}$ (one control 0.125), halving whenever a finer level appears; $\kappa_2 = 0$; $1 + \log$ slicing and Gamma-driver shift with advection terms, $\mu_S = 3/4$, $\eta = 2$; Kreiss–Oliger dissipation 0.5; floor 10^{-5} on χ; four ghost cells (two in the test-particle meshes).

[b] Löhner indicator on $q = \alpha\chi^{-7/4}$: refine above 0.2, derefine below 0.02, $\epsilon_0 = 0.01$, $\delta_0 = 10^{-99}$; checked every 4 cycles, regrids at least 8 cycles apart.

[c] Searches every 16 cycles on 16 polar points up to degree 8, stopping tolerance 10^{-5} on the mass, at most 400 flow iterations, flow constant 1; the stopping-rule control: every 160 cycles, 10^{-8}, 4000 iterations.

TABLE III. Measured quantities to which the text refers for test particles and the apparent-horizon check, with the values of Ref. [21] where it gives them (from its text, or read from its figures). Masses in units of the total mass M, times in units of M; listed values follow the order named in the first column.

Quantity	This work	Ref. [21]
Fixed black hole		
Passage $e = 0.99$, largest relative errors of $\mathcal{E}$, L_z, Q ($\Delta t = 0.01M$, to $230M$):		
implicit midpoint, analytic metric	2.78×10^{-8}, 0, 1.25×10^{-7}	2.8×10^{-8}, round-off, 1.3×10^{-7}
spacetime Boris, symmetric gauge, analytic	9.05×10^{-8}, 1.14×10^{-8}, 7.83×10^{-7}	9.0×10^{-8}, 1.1×10^{-8}, 7.8×10^{-7}
implicit midpoint, Hermite lattice	2.16×10^{-6}, 5.30×10^{-6}, 5.06×10^{-5}	2.2×10^{-6}, 5.3×10^{-6}, 5.1×10^{-5}
spacetime Boris, Cholesky gauge, Hermite lattice	2.05×10^{-6}, 3.79×10^{-6}, 5.29×10^{-5}	2.0×10^{-6}, 3.8×10^{-6}, 5.3×10^{-5}
Passage, order of convergence in Δt (three pushers, analytic metric)	1.995–2.001	second order
Inclined Kerr orbit on the mesh, order in h of position and $\mathcal{E}$ errors	2.89, 3.29	–
Plunging clusters ($N = 10^4$): removed by excision, against exact geodesics; $a \to 0$ and $a = 0.9M$	3726 vs 3727; 3345 vs 3339	–
Bound clusters, exponent of the filtered-E differences against N; $a \to 0$ and $a = 0.9M$	-0.48, -0.48	–
Evolving black holes		
Puncture: largest relative change of L_z of the four circular tracers	1.1×10^{-4}–3.9×10^{-4}	–
Puncture: radial infall and plunge removed at $\alpha_{\rm exc}$, t/M	54, 126	–
Puncture ($\Delta x = M/8$), slice Misner–Sharp horizon mass at $t = 50M$	1.0016	–
Head-on: first common apparent horizon t/M; its mass at the end	49.25; 1.0026	–
Head-on: largest relative change of L_x (five orbiting tracers); $\lvert L_x \rvert$ of the tracer from rest	9.8×10^{-4}; 4×10^{-14}	–
Apparent-horizon check		
Finder as distributed, $t = 50M$, at $\Delta x = M/4$, $M/8$, $M/16$	0.9353, 0.9771, 0.9406	–
Slice Misner–Sharp horizon mass, same slices	0.9656, 1.0016, 1.0000	–
Finder as distributed ($M/8$), stopping tolerance 10^{-5} and 10^{-8}, $t \simeq 50M$	0.9771, 0.9771	–
Corrected finder in a live run ($M/8$), last search; slice mass at $t = 12M$	1.0003; 1.0006	–

tures evaluate Eq. (37) directly, while spherical Gaussian components of conformal mass m_c and width σ_c obey

$$\tilde{\rho}_c(\bar{r}_c) = \frac{m_c}{(2\pi\sigma_c^2)^{3/2}} \exp\!\left(-\frac{\bar{r}_c^2}{2\sigma_c^2}\right), \qquad u_c(\bar{r}_c) = \frac{m_c}{2\bar{r}_c} \operatorname{erf}\!\left(\frac{\bar{r}_c}{\sqrt{2}\,\sigma_c}\right). \tag{B1}$$

A homogeneous prolate spheroid of ADM mass M, polar semi-axis b along z, eccentricity e, equatorial semi-axis $a_{\rm eq} = b\sqrt{1-e^2}$, and focal half-distance $d_{\rm f} = eb$ has closed-form fields [54, 56]

$$\begin{aligned}
\tilde{\rho}_c(\boldsymbol{x}) &= \frac{3M}{4\pi a_{\rm eq}^2 b}, \qquad \frac{x^2+y^2}{a_{\rm eq}^2} + \frac{z^2}{b^2} \le 1, \\
u_c(\boldsymbol{x}) &= \frac{3M}{8}\Big[I_0(\lambda) - (x^2+y^2)J_{\rm eq}(\lambda) - z^2 J_z(\lambda)\Big], \\
I_0(\lambda) &= \frac{2}{d_{\rm f}} \operatorname{arctanh}\!\left(\frac{d_{\rm f}}{b_\lambda}\right), \\
J_{\rm eq}(\lambda) &= \frac{b_\lambda}{d_{\rm f}^2(a_{\rm eq}^2+\lambda)} - \frac{I_0(\lambda)}{2d_{\rm f}^2}, \\
J_z(\lambda) &= \frac{I_0(\lambda)}{d_{\rm f}^2} - \frac{2}{d_{\rm f}^2 b_\lambda}, \\
1 &= \frac{x^2+y^2}{a_{\rm eq}^2+\lambda} + \frac{z^2}{b^2+\lambda} \qquad (\lambda > 0).
\end{aligned} \tag{B2}$$

The auxiliary length $b_\lambda = \sqrt{b^2+\lambda}$ sets $\lambda = 0$ inside the spheroid, while outside the boundary $\tilde{\rho}_c$ vanishes and $\lambda > 0$ solves the last relation. Every initial slice sets vanishing

TABLE IV. As Table III, for dust collapse. Hamiltonian constraint norm $\|\mathcal{H}\| = (\int \mathcal{H}^2 \sqrt{\gamma}\, \mathrm{d}^3x)^{1/2}$, integrated over the grid cells with $\chi \geq 1/16$ (a total, without division by a volume); the same measure in every table.

Quantity	This work	Ref. [21]
Dust collapse on a lattice		
Central proper time τ_c/M at which α_c exceeds the lapse bound by 10^{-2}	3.53	$\simeq 3.5$
Rest mass $\sum_p m_p/M$: published lattice rule; surface-corrected (continuum M_0)	1.1687; 1.1551	$\simeq 1.155$ ($N \simeq 10^6$)
First apparent horizon t/M (finest-level clouds, corrected masses)	16.39	–
M_{AH}/M at formation; at $t = 60M$ (finest-level clouds, corrected masses)	1.0004; 0.9987	$\lvert M_{\mathrm{AH}}/M - 1\rvert \lesssim 10^{-3}$ at $60M$
M_{AH}/M at formation; at $t = 60M$ (published lattice masses)	1.0114; 1.0099	–
M_{AH}/M when the last macroparticle is removed → at $t = 60M$: finest level; level cap with any, ≤ 2, ≤ 1 levels of coarsening	1.0004→0.9987; 1.0006→1.0065; 1.0006→1.0064; 1.0012→0.9904	–
Largest $\|\mathcal{H}\|$, same four runs	1.9, 4.7, 4.5, 2.8	–
Largest $\|\mathcal{H}\|$ within $1M$ after the refinement to $\Delta x = h/4$, same four runs	1.59, 0.48, 0.48, 1.14	–
Central reading of E (eight central cells, after the filter) over the cycloid at the same τ_c, in the $3M$ after the refinement to $h/4$: finest level; level cap with any, ≤ 1 levels of coarsening	0.27–0.52; 1.11–1.14; 1.28–1.49	–
Published lattice masses: τ_c/M at which α_c exceeds the lapse bound by 10^{-2}; central reading of E over the cycloid before the refinement to $h/4$; largest $\lvert\Delta\alpha_c\rvert$ from the corrected-mass run while macroparticles remain	3.56; 0.945–1.002; 4.7×10^{-3}	–
Dust collapse from a random sample		
Random sample, order of the runs below: capped clouds at the full depth with $n_{\mathrm{f}} = 1$ and 4+c; one level fewer, finest-level and capped clouds (4+c)		
first apparent horizon t/M, M_{AH}/M	16.35, 0.9949; 16.31, 0.9939; 16.20, 1.0019; 16.58, 0.9990	–
M_{AH}/M at $t = 20M$, $40M$, and over the last $2M$ (last search t/M); medians over $\pm 1M$, as successive searches scatter by up to 10^{-3}	1.0000, 1.0004, 1.0000 (43); 1.0004, 1.0013, 1.0013 (41); 0.9983, 0.9971, 0.9973 (60); 0.9984, 0.9946, 0.9946 (60)	–
Largest MeshBlock count: the four runs above; finest-level clouds at the full depth with $n_{\mathrm{f}} = 1$ and 4+c, count asked for when the run stopped (t/M)	1779, 1681, 400, 400; 1884 (8.8), 2024 (10.2)	–
One level fewer, largest $\|\mathcal{H}\|$ before the first horizon, from it until the last macroparticle is removed, and afterwards: finest-level; capped clouds	1.77, 2.93, 0.91; 2.95, 3.79, 2.23	–

shift $\beta^i = 0$, with initial lapse $\alpha = 1$ for cold collapse and $\alpha = \psi^{-2}$ for vacuum punctures. Isolated clusters use their static equilibrium lapse joined outside to $(1-u_c)(1+u_c)^{-1}$, whereas the composite cluster uses $\alpha = 2\psi^{-1} - 1$. Since $\sqrt{\gamma} = \psi^6$, N equal-mass macroparticles sampling proper rest-mass density ρ_0 and continuum rest mass M_0 have coordinate number density

$$\mathcal{N}(\boldsymbol{x}) = \frac{N}{M_0}\, \rho_0(\boldsymbol{x}) \psi(\boldsymbol{x})^6, \qquad m_p = \frac{M_0}{N}, \qquad M_0 = \int \rho_0(\boldsymbol{x}) \psi(\boldsymbol{x})^6\, \mathrm{d}^3x. \tag{B3}$$

For the cold prolate spheroid ($u_{i,p} = 0$, $\rho_0 \psi^6 = \tilde{\rho}_c \psi$), one quarter of the N particles are drawn inside the boundary by rejection against $\psi(\boldsymbol{x})\psi(\boldsymbol{0})^{-1}$ and completed by $(x, y, -z)$ and $(-x, -y, \pm z)$. Gaussian components draw positions around $\boldsymbol{x}_c$ by randomized stratification in radius and angles. In multi-component superpositions, a tabulated equilibrium envelope multiplies its isolated particle masses by $\psi\psi_c^{-1}$ and lowers its isolated orthonormal momenta with ψ^2. Each Gaussian component of N_c particles at bulk Lorentz factor W_c sets $m_p = m_c N_c^{-1} \psi W_c^{-1}$ and lowers its bulk orthonormal momentum with ψ^2.

The homogeneous dust ball of mass M, areal radius R_0, and density ρ_0 (Sec. V A) starts from rest ($u_{i,p} = 0$) with isotropic radius $\bar{r}_0$ and interior conformal factor [53]

$$\bar{r}_0 = \frac{R_0}{2}\left(1 - \frac{M}{R_0} + \sqrt{1 - \frac{2M}{R_0}}\right), \qquad \psi(\bar{r}) = \sqrt{\frac{\left(1 + \sqrt{1 - 2M/R_0}\right)\bar{r}_0 R_0^2}{2\bar{r}_0^3 + M\bar{r}^2}} \qquad (\bar{r} \leq \bar{r}_0), \tag{B4}$$

matched to Eq. (37) outside. The cube $[-\bar{r}_0, \bar{r}_0]^3$ is

TABLE V. As Table III, for the collapse of a prolate spheroid; $\|\mathcal{H}\|$ as in Table IV.

Quantity	This work	Ref. [21]
Prolate collapse		
Largest Kretschmann scalar: time t/M and $\|z\|/M$ (on the axis)	27.75; 0.66	–
First apparent horizon t/M; $M_{\rm AH}/M$	32.00; 0.929	32 (from its text)
Largest $M_{\rm AH}/M$ (t/M); $M_{\rm AH}/M$ at the last search (t/M)	0.9666 (33.75); 0.9489 (44.75)	–
First and last removal of macroparticles t/M (macroparticles left)	33.78; 38.80 (0)	–
Misner–Sharp mass on the last slice, at the horizon ($\pm$ its spread over angle) and at large radius (t/M of the slice)	0.920 $\pm$ 0.027; 0.999 (45)	–
Largest $\|\mathcal{H}\|$ (t/M); lattice dust collapse (finest level, corrected masses) on the same measure	6.31 (27.5); 1.90	–

TABLE VI. As Table III, for the relativistic clusters; $\|\mathcal{H}\|$ as in Table IV. Central lapse α_c in the eight central cells; $\alpha_c(0)$ is its value at $t = 0$.

Quantity	This work	Ref. [21]
Stable cluster $y_c = 0.819$: filter		
Rise of α_c above $\alpha_c(0)$ within $130M$, sample 1, $n_{\rm f} = 0$, 2, 4, 2+c, 4+c	0.0044, 0.0176, 0.0273, 0.0061, 0.0065	≈ 0.01 (kicked model; read from figure)
Same rise, samples 1–3, four plain passes; with compensation	0.0273, 0.0289, 0.0275; 0.0065, 0.0084, 0.0061	–
Filtered over deposited proper-volume integral of E, minus one: four plain passes; with compensation	-3.0×10^{-3}; -1.0×10^{-4}	–
$\|\mathcal{H}\|$ at $t = 300M$: no filter; four plain passes; with compensation	0.141; 0.011; 0.031	–
Sample 1, four plain passes; with compensation: largest radius enclosing 50 and 90 percent of the rest mass over its initial value; lowest central E ($20M$ means) over its first value	1.16, 1.16, 0.66; 1.03, 1.03, 0.84	–
Stable cluster $y_c = 0.819$: core oscillation		
Period of α_c (100–1200M); radial period in the core, $\pi[3/(4\pi E_c)]^{1/2}$	75; 66	–
Half the swing of α_c in windows of $100M$: growth phase (100–500M); after $500M$	0.005→0.018; 0.015–0.022	–
Largest $\|\Delta\alpha_c\|$ from sample 1 over $300M$: $\Delta t/2$, $\kappa_1 = 0.1$, $\kappa_1 = 0$, c_H	6.5×10^{-4}, 3.1×10^{-4}, 7.9×10^{-5}, 2.1×10^{-5}	–
Same, compensation that also cancels the cloud width	3.6×10^{-3}	–
Sample 1: range of the radius enclosing 10 and of that enclosing 90 percent of the rest mass over its initial value, 0–$300M$; 300–$1200M$	0.18, 0.07; 0.51, 0.25	–
Half the cell spacing over the cluster against sample 1 (to $300M$): first maximum of $\alpha_c - \alpha_c(0)$ (t/M), growth rate of the half-swing per M, last minimum (t/M), mean of the last two extrema; sample 1 first	+0.0037 (24), 1.3×10^{-5}, −0.0108 (280), −0.0039; +0.0038 (23), 1.7×10^{-5}, −0.0155 (276), −0.0071	–
Near-critical $y_c = 0.756$ in Ref. [21]: period of α_c; here the core radial period of that model	45	≈ 50 (read from figure)
Stable cluster $y_c = 0.819$: particle number		
Monte Carlo, $N = 10^3$ against 10^4: largest difference of $\alpha_c - \alpha_c(0)$ to $190M$; lowest $\alpha_c - \alpha_c(0)$ after $190M$ (10^4, 10^3); mean over 400–$500M$ (10^4, 10^3)	0.005; −0.023, −0.035; −0.003, −0.017	–
Monte Carlo $N = 10^2$: α_c at $t = 0$, $200M$, $500M$; mean Lorentz factor at start and end; largest $\|\sum_p m_p u_{i,p}\|/M_0$	0.81, 0.68, 0.58; 1.047, 1.138; 0.039	–
Cluster sequence with inward kick		
α_c range over $300M$, $y_c = 0.900$; 0.819; particles kept	0.886–0.892; 0.787–0.819; all	≈ 0.89; 0.81–0.82 (read from figure)
Largest change of the mean Lorentz factor $\sum_p m_p W_p / \sum_p m_p$ over $300M$, $y_c = 0.900$; 0.819	2.1×10^{-4}; 1.5×10^{-3}	–
First apparent horizon t/M: $y_c = 0.605$; 0.500	191.0; 104.3	109; 73
$M_{\rm AH}/M$ at the end; macroparticles left of 10^4	1.0000; 1.0001; 6, 3	$\simeq 1$ (stated)
α_c at the end, after the horizon forms: $y_c = 0.605$; 0.500	0.051; 0.016	≈ 0.20 (read from figure)
$y_c = 0.283$: α_c at the end (t/M); particles kept	0.976 (270); all	≈ 0.98 (300; read from figure)

TABLE VII. As Table VI, for the unperturbed model past the binding-energy maximum and the near-critical model.

Quantity	This work	Ref. [21]
Unperturbed $y_c = 0.5$		
Time t/M at which α_c first reaches 0.8: plain filter; compensated, samples 1–3	59; 88, 86, 91	$\approx$ 120 (read from figure)
Migration: α_c at the start and at its extrema to $1000M$ (t/M)	0.552 (0); 0.913 (228); 0.591 (400); 0.637 (417); 0.541 (436); 0.894 (631); 0.753 (783); 0.853 (940)	0.56 (0); 0.905 (200–320); 0.55–0.63 (440–520); 0.90 (670); 0.765 (830–880) (read from figure)
Migration: extrema after $1000M$ (t/M); α_c at the end; mean over 1500–2500M	0.795 (1072); 0.853 (1184); 0.792 (1329); 0.840 (1472); 0.802 (1582); 0.833 (1726); 0.804 (1852); 0.826 (2001); 0.816 (2500); 0.814	0.865 (1040); 0.78 (1160); 0.865 (1320); 0.80 (1450); 0.85 (1600) (read from figure); late mean 0.834 (from its text)
Migration: macroparticles removed by excision; leaving the domain (at t/M)	0; 4 (1816, 1999, 2405, 2440)	–
Near-critical $y_c = 0.756$		
Compressed by ten percent: $\alpha_c(0)$; first apparent horizon t/M	0.749; 147.2	$\simeq$ 0.75; none
Compressed: $M_{\rm AH}/M$ at the end; M of the initial data; macroparticles left	0.9970; 0.9969; 0	all kept
Compressed: largest $\lvert\Delta\alpha_c\rvert$ to $120M$ between the consistent and the unperturbed initial metric	0.009	–
Density lowered by ten percent: α_c range; mean over $600M$; radii enclosing 90 and 10 percent of the rest mass, range $(t = 0)$; macroparticles kept	0.721–0.827; 0.783; 5.8–9.2 (6.4), 1.8–3.8 (2.4); all	–
Unperturbed: spacing of successive minima of α_c from $200M$, before and after $600M$	44–58; 50–76	maxima 50–55 to 360, 60–70 after (perturbed run; read from figure)
Unperturbed: largest half-swing of α_c between successive extrema (t/M)	0.036 (415)	0.041 (250–275; perturbed run; read from figure)
Unperturbed: α_c range; mean of the last two pairs of successive extrema (t/M); macroparticles kept	0.696–0.772; 0.720 (745), 0.714 (770); all	0.696–0.777; 0.735 (800; perturbed run; read from figure); all kept

partitioned into $n_{\rm lat}^3$ cubic cells ($n_{\rm lat} = 26$) of spacing $h = 2\bar{r}_0 n_{\rm lat}^{-1}$ aligned with the initial matter-level mesh, with h and site count N in Table I. Macroparticles occupy cell centers with $\bar{r}_p = |\boldsymbol{x}_p| \le \bar{r}_0$ for noise-free interior deposition, and published lattice masses [21] assign proper cell masses

$$m_p^{\rm pub} = \rho_0 \psi(\bar{r}_p)^6 h^3. \tag{B5}$$

The count of lattice sites inside a sphere differs from its volume in cells h^3 by a surface fraction of order $n_{\rm lat}^{-1}$ of either sign, and at $n_{\rm lat} = 26$ summed $m_p^{\rm pub}$ exceeds M_0 (Table IV). Our surface correction leaves interior masses unchanged for $\bar{r}_p \le \bar{r}_0 - h$ and rescales the outermost layer $\bar{r}_0 - h < \bar{r}_p \le \bar{r}_0$ of width h by one common factor so all particle masses sum to M_0,

$$m_p = m_p^{\rm pub} \frac{M_0 - \sum_{\bar{r}_{p'} \le \bar{r}_0 - h} m_{p'}^{\rm pub}}{\sum_{\bar{r}_{p'} > \bar{r}_0 - h} m_{p'}^{\rm pub}}. \tag{B6}$$

Monoenergetic equilibrium clusters [1] populate a single conserved Killing energy $\mathcal{E} = \alpha_{\rm s}$ (Sec. V C) with isotropic momentum directions. In dimensionless areal radius $\tilde{r}$, squared lapse ratio $y(\tilde{r})$ with $y(0) = y_c$, and enclosed gravitational and rest-mass variables $\tilde{m}$ and $\tilde{m}_0$, the structure equations and isotropic-radius map take the form

$$\begin{aligned}
\zeta &= \frac{1 - y_c}{3}, \qquad g_y = \frac{y^{-1} - 1}{y_c^{-1} - 1}, \\
\frac{{\rm d}y}{{\rm d}\tilde{r}} &= \frac{2\zeta y}{\tilde{r}^2} \frac{\tilde{m} + \zeta \tilde{r}^3 (y/y_c)^{-1/2} g_y^{3/2}}{1 - 2\zeta\tilde{m}/\tilde{r}}, \\
\frac{{\rm d}\tilde{m}}{{\rm d}\tilde{r}} &= \tilde{r}^2 \left(\frac{y}{y_c}\right)^{-3/2} g_y^{1/2}, \\
\frac{{\rm d}\tilde{m}_0}{{\rm d}\tilde{r}} &= \frac{\tilde{r}^2 y_c^{1/2} (y/y_c)^{-1} g_y^{1/2}}{\sqrt{1 - 2\zeta\tilde{m}/\tilde{r}}}, \\
\frac{{\rm d}\ln\bar{r}}{{\rm d}\ln\tilde{r}} &= \frac{1}{\sqrt{1 - 2\zeta\tilde{m}/\tilde{r}}}.
\end{aligned} \tag{B7}$$

Near the cluster surface where density vanishes, the variable $\sqrt{1-y}$ keeps every derivative analytic up to $y(\tilde{r}_{\rm s}) = 1$. Outward integration from $\tilde{r} = 0$ ($\tilde{m} = \tilde{m}_0 = 0$) yields surface values $(\tilde{r}_{\rm s}, \tilde{m}_{\rm s}, \tilde{m}_{0,{\rm s}})$ and interior profiles matched to Eq. (37) at the isotropic surface radius $\bar{r}_{\rm s}$ given by the first line of Eq. (B4) for R,

$$\begin{aligned}
R &= \frac{M\tilde{r}_{\rm s}}{\zeta\tilde{m}_{\rm s}}, \qquad M_0 = \frac{M\tilde{m}_{0,{\rm s}}}{\tilde{m}_{\rm s}}, \\
\alpha_{\rm s} &= \sqrt{1 - \frac{2M}{R}}, \qquad \alpha_{\rm eq}(\bar{r}) = \alpha_{\rm s}\sqrt{y}, \\
\psi_{\rm eq}(\bar{r}) &= \sqrt{\frac{R\tilde{r}}{\tilde{r}_{\rm s}\bar{r}}}.
\end{aligned} \tag{B8}$$

Enclosed gravitational and rest masses follow from $m_{\rm g} = M\tilde{m}\tilde{m}_{\rm s}^{-1}$ and $m_0 = M_0\tilde{m}_0\tilde{m}_{0,{\rm s}}^{-1}$, with $4\pi\bar{r}^2\rho_0\psi_{\rm eq}^6\,{\rm d}\bar{r} = {\rm d}m_0$. For N equal-mass macroparticles ($p = 1, \ldots, N$) and five coordinates $d \in \{1, \ldots, 5\}$, randomized stratification draws

$$U_{pd} = \frac{\pi_d(p) + \xi_{pd}}{N}, \tag{B9}$$

where independent random permutations π_d map $\{1, \ldots, N\}$ onto $\{0, \ldots, N-1\}$ and ξ_{pd} is uniform on $[0, 1)$. The cumulative rest-mass fraction $\mathcal{F}(\bar{r}) = m_0(\bar{r})M_0^{-1}$ sets each radial coordinate through $\bar{r}_p = \mathcal{F}^{-1}(U_{p1})$. Position angles $(\cos\theta_p, \phi_p) = (2U_{p2} - 1, 2\pi U_{p3})$ define the radial unit vector $\hat{\boldsymbol{r}}_p$ and $\boldsymbol{x}_p = \bar{r}_p\hat{\boldsymbol{r}}_p$, while momentum angles $(\cos\theta_{u,p}, \phi_{u,p}) = (2U_{p4} - 1, 2\pi U_{p5})$ define an isotropic unit vector $\hat{\boldsymbol{u}}_p$. At radius $\bar{r}_p$, the monoenergetic condition $\mathcal{E} = \alpha_{\rm eq}(\bar{r}_p)W_p = \alpha_{\rm s}$ and Eq. (5) fix the local orthonormal momentum magnitude $u_{\rm eq}(\bar{r}_p)$ and covariant momentum,

$$\begin{aligned}
u_{\rm eq}(\bar{r}) &= \sqrt{\frac{\alpha_{\rm s}^2}{\alpha_{\rm eq}(\bar{r})^2} - 1}, \\
u_{i,p} &= \psi_{\rm eq}(\bar{r}_p)^2 u_{\rm eq}(\bar{r}_p)\hat{u}_{i,p}.
\end{aligned} \tag{B10}$$

Plain Monte Carlo sampling, used as the alternative in the particle-number study, replaces U_{pd} in Eq. (B9) with independent uniform draws on $[0, 1)$. Because the mass-weighted mean momentum of a finite sample does not vanish and makes the cluster drift, that mean is subtracted from every particle,

$$u_{i,p} \leftarrow u_{i,p} - \frac{\sum_{p'=1}^N m_{p'} u_{i,p'}}{\sum_{p'=1}^N m_{p'}}. \tag{B11}$$

Along the equilibrium sequence, a coherent inward velocity kick of amplitude δ_v, consistent with the description in Ref. [21], adds an inward radial component by replacing $\hat{u}_{i,p}$ in Eq. (B10) with $\hat{u}_{i,p} - \delta_v\hat{r}_{i,p}$ before Eq. (B11). For the homologous density perturbation δ_ρ of the near-critical model, equilibrium radii $\bar{r}_{{\rm eq},p}$ scale by a common factor $\Lambda_{\rm h}$ to $\bar{r}_p = \Lambda_{\rm h}\bar{r}_{{\rm eq},p}$ with $\boldsymbol{x}_p = \bar{r}_p\hat{\boldsymbol{r}}_p$. Unlike a perturbation applied on the unperturbed metric $\psi_{\rm eq}$, here $\psi(\bar{r})$ is recomputed from the perturbed matter by a radial fixed-point solve of the Hamiltonian constraint,

$$\begin{aligned}
\Lambda_{\rm h} &= (1 + \delta_\rho)^{-1/3}, \\
\frac{1}{\bar{r}^2}\frac{\rm d}{{\rm d}\bar{r}}\left(\bar{r}^2\frac{{\rm d}\psi}{{\rm d}\bar{r}}\right) &= -\frac{2\pi}{\Lambda_{\rm h}^3\psi(\bar{r})}\rho_0\left(\frac{\bar{r}}{\Lambda_{\rm h}}\right)\psi_{\rm eq}\left(\frac{\bar{r}}{\Lambda_{\rm h}}\right)^6\langle W(\bar{r})\rangle.
\end{aligned} \tag{B12}$$

Outside $\Lambda_{\rm h}\bar{r}_{\rm s}$, $\psi(\bar{r})$ joins $1 + M'(2\bar{r})^{-1}$ for perturbed ADM mass M', while the interior lapse rescales $\alpha_{\rm eq}(\bar{r}\Lambda_{\rm h}^{-1})$ continuously to the exterior isotropic Schwarzschild lapse. Particle momenta at $\bar{r}_p$ are set from $\mathcal{E} = \alpha_{\rm s}$ via $u_{\rm eq}(\bar{r}_{{\rm eq},p})$ lowered with $\psi_{\rm eq}(\bar{r}_p)^2$ before Eq. (B11), while $\langle W(\bar{r})\rangle$ averages Eq. (5) over directions in the new metric $\psi(\bar{r})^4\delta_{ij}$. An inward velocity kick with vanishing extrinsic curvature $K_{ij} = 0$ leaves a first-order

violation of the momentum constraint $\mathcal{M}_i = 0$, which is not corrected.

The relativistic cluster with two collapsing overdensities superposes a monoenergetic envelope and two half-turn-symmetric cold Gaussian components on an equatorial circular orbit of radius $d_{\rm orb}$. At orbital radius $d_{\rm orb}$ enclosing envelope mass $m_{\rm g}(d_{\rm orb})$, leading-order circular motion gives

$$v_{\rm orb}^2 = \frac{m_{\rm g}(d_{\rm orb})}{d_{\rm orb}} + \frac{m_c}{4d_{\rm orb}}, \tag{B13}$$

which matches the speed quoted in Ref. [21] only for a host cluster less than half as large as the stated radius. The cluster size in the published figure agrees with that smaller value, and our initial-data construction adopts this self-consistent host radius.

[1] S. L. Shapiro and S. A. Teukolsky, Relativistic stellar dynamics on the computer. I - Motivation and numerical method, ApJ **298**, 34 (1985).
[2] J. Binney and S. Tremaine, *Galactic Dynamics: Second Edition* (2008).
[3] H. Andreasson, The Einstein-Vlasov System/Kinetic Theory, Living Rev. Rel. **14**, 4 (2011), arXiv:1106.1367 [gr-qc].
[4] R. E. Angulo and O. Hahn, Large-scale dark matter simulations, Liv. Rev. Comput. Astrophys. **8**, 1 (2022), arXiv:2112.05165 [astro-ph.CO].
[5] J. Bamber, S. L. Shapiro, M. Ruiz, and A. Tsokaros, Evolution of a black hole cluster in full general relativity, Phys. Rev. D **112**, 024046 (2025), arXiv:2505.01495 [gr-qc].
[6] C. K. Birdsall and A. B. Langdon, *Plasma Physics via Computer Simulation* (1991).
[7] W. E. East, R. Wojtak, and F. Pretorius, Einstein-Vlasov Calculations of Structure Formation, Phys. Rev. D **100**, 103533 (2019), arXiv:1908.05683 [astro-ph.CO].
[8] O. Sarbach and T. Zannias, Relativistic Kinetic Theory: An Introduction, AIP Conf. Proc. **1548**, 134 (2013), arXiv:1303.2899 [gr-qc].
[9] R. O. Acuña-Cárdenas, C. Gabarrete, and O. Sarbach, An introduction to the relativistic kinetic theory on curved spacetimes, Gen. Rel. Grav. **54**, 23 (2022), arXiv:2106.09235 [gr-qc].
[10] J. R. Ipser and K. S. Thorne, Relativistic, Spherically Symmetric Star Clusters. I. Stability Theory for Radial Perturbations, ApJ **154**, 251 (1968).
[11] J. R. Ipser, A binding-energy criterion for the dynamical stability of spherical stellar systems in general relativity, ApJ **238**, 1101 (1980).
[12] S. L. Shapiro and S. A. Teukolsky, Relativistic Stellar Dynamics on the Computer - Part Two - Physical Applications, ApJ **298**, 58 (1985).
[13] G. Rein, A. D. Rendall, and J. Schaeffer, Critical collapse of collisionless matter: A Numerical investigation, Phys. Rev. D **58**, 044007 (1998), arXiv:gr-qc/9804040.
[14] I. Olabarrieta and M. W. Choptuik, Critical phenomena at the threshold of black hole formation for collisionless matter in spherical symmetry, Phys. Rev. D **65**, 024007 (2002), arXiv:gr-qc/0107076.
[15] A. Akbarian and M. W. Choptuik, Critical collapse in the spherically-symmetric Einstein-Vlasov model, Phys. Rev. D **90**, 104023 (2014), arXiv:1409.5176 [gr-qc].
[16] A. M. Abrahams, G. B. Cook, S. L. Shapiro, and S. A. Teukolsky, Solving Einstein's equations for rotating space-times: Evolution of relativistic star clusters, Phys. Rev. D **49**, 5153 (1994).
[17] M. Shibata, 3-D numerical simulation of black hole formation using collisionless particles: Triplane symmetric case, Prog. Theor. Phys. **101**, 251 (1999).
[18] M. Shibata, Fully general relativistic simulation of merging binary clusters: Spatial gauge condition, Prog. Theor. Phys. **101**, 1199 (1999), arXiv:gr-qc/9905058.
[19] W. E. East, Cosmic Censorship Upheld in Spheroidal Collapse of Collisionless Matter, Phys. Rev. Lett. **122**, 231103 (2019), arXiv:1901.04498 [gr-qc].
[20] E. Ames, H. Andréasson, and O. Rinne, Dynamics of gravitational collapse in the axisymmetric Einstein–Vlasov system, Class. Quant. Grav. **38**, 105003 (2021), arXiv:2010.15771 [gr-qc].
[21] J. Wu and E. R. Most, Numerical-Relativity Particle-in-Cell Simulations: Self-Gravitating Collisionless Matter in Dynamical Spacetimes (2026), arXiv:2611.12831.
[22] R. W. Hockney and J. W. Eastwood, *Computer simulation using particles* (1988).
[23] A. Spitkovsky, Simulations of relativistic collisionless shocks: shock structure and particle acceleration, AIP Conf. Proc. **801**, 345 (2005), arXiv:astro-ph/0603211.
[24] B. Cerutti, G. R. Werner, D. A. Uzdensky, and M. C. Begelman, Simulations of particle acceleration beyond the classical synchrotron burnoff limit in magnetic reconnection: An explanation of the Crab flares, Astrophys. J. **770**, 147 (2013), arXiv:1302.6247 [astro-ph.HE].
[25] H. Hakobyan *et al.* (Entity Development Team), Entity—Hardware-agnostic Particle-in-cell Code for Plasma Astrophysics. I. Curvilinear Special Relativistic Module, Astrophys. J. Supp. **285**, 11 (2026), arXiv:2511.17710 [astro-ph.HE].
[26] J. P. Boris *et al.*, Relativistic plasma simulation-optimization of a hybrid code, in *Proc. Fourth Conf. Num. Sim. Plasmas* (1970) pp. 3–67.
[27] F. Bacchini, B. Ripperda, A. A. Philippov, and K. Parfrey, A Coupled Guiding Center–Boris Particle Pusher for Magnetized Plasmas in Compact-object Magnetospheres, Astrophys. J. Suppl. **251**, 10 (2020), arXiv:2007.14401 [astro-ph.HE].
[28] H. Olivares, I. M. Peshkov, E. R. Most, F. M. Guercilena, and L. J. Papenfort, New first-order formulation of the Einstein equations exploiting analogies with electrodynamics, Phys. Rev. D **105**, 124038 (2022), arXiv:2111.05282 [gr-qc].
[29] S. Boyeneni, J. Wu, and E. R. Most, Unveiling the Electrodynamic Nature of Spacetime Collisions, Phys. Rev. Lett. **135**, 101401 (2025), arXiv:2504.15978 [gr-qc].
[30] A. Myers *et al.*, Porting WarpX to GPU-accelerated

platforms, Parallel Comput. **108**, 102833 (2021), arXiv:2101.12149 [physics.comp-ph].
[31] C. Bona, T. Ledvinka, C. Palenzuela, and M. Zacek, General covariant evolution formalism for numerical relativity, Phys. Rev. D **67**, 104005 (2003), arXiv:gr-qc/0302083.
[32] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, Accurate evolutions of orbiting black-hole binaries without excision, Phys. Rev. Lett. **96**, 111101 (2006), arXiv:gr-qc/0511048.
[33] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, Gravitational wave extraction from an inspiraling configuration of merging black holes, Phys. Rev. Lett. **96**, 111102 (2006), arXiv:gr-qc/0511103.
[34] D. Hilditch, S. Bernuzzi, M. Thierfelder, Z. Cao, W. Tichy, and B. Bruegmann, Compact binary evolutions with the Z4c formulation, Phys. Rev. D **88**, 084057 (2013), arXiv:1212.2901 [gr-qc].
[35] J. M. Stone, P. D. Mullen, D. Fielding, P. Grete, M. Guo, P. Kempski, E. R. Most, C. J. White, and G. N. Wong, AthenaK: A Performance-portable Version of the Athena++ Adaptive Mesh Refinement Framework, ApJS **283**, 27 (2026), arXiv:2409.16053 [astro-ph.IM].
[36] H. Zhu, J. Fields, F. Zappa, D. Radice, J. M. Stone, A. Rashti, W. Cook, S. Bernuzzi, and B. Daszuta, Performance-portable Numerical Relativity with AthenaK, Astrophys. J. Suppl. **278**, 50 (2025), arXiv:2409.10383 [gr-qc].
[37] C. R. Trott, D. Lebrun-Grandié, D. Arndt, J. Ciesko, V. Dang, N. Ellingwood, R. Gayatri, E. Harvey, D. S. Hollman, D. Ibanez, N. Liber, J. Madsen, J. Miles, D. Poliakoff, A. Powell, S. Rajamanickam, M. Simberg, D. Sunderland, B. Turcksin, and J. Wilke, Kokkos 3: Programming Model Extensions for the Exascale Era, ITPDS **33**, 805 (2022).
[38] J. R. Oppenheimer and H. Snyder, On Continued gravitational contraction, Phys. Rev. **56**, 455 (1939).
[39] J. A. Font, T. Goodale, S. Iyer, M. A. Miller, L. Rezzolla, E. Seidel, N. Stergioulas, W.-M. Suen, and M. Tobias, Three-dimensional general relativistic hydrodynamics. 2. Long term dynamics of single relativistic stars, Phys. Rev. D **65**, 084024 (2002), arXiv:gr-qc/0110047.
[40] R. L. Arnowitt, S. Deser, and C. W. Misner, The Dynamics of general relativity, Gen. Rel. Grav. **40**, 1997 (2008), arXiv:gr-qc/0405109.
[41] C. Gundlach, J. M. Martin-Garcia, G. Calabrese, and I. Hinder, Constraint damping in the Z4 formulation and harmonic gauge, Class. Quant. Grav. **22**, 3767 (2005), arXiv:gr-qc/0504114.
[42] M. Alcubierre, B. Bruegmann, P. Diener, M. Koppitz, D. Pollney, E. Seidel, and R. Takahashi, Gauge conditions for long term numerical black hole evolutions without excision, Phys. Rev. D **67**, 084023 (2003), arXiv:gr-qc/0206072.
[43] J. R. van Meter, J. G. Baker, M. Koppitz, and D.-I. Choi, How to move a black hole without excision: Gauge conditions for the numerical evolution of a moving puncture, Phys. Rev. D **73**, 124011 (2006), arXiv:gr-qc/0605030.
[44] C. Bona, J. Masso, E. Seidel, and J. Stela, A New formalism for numerical relativity, Phys. Rev. Lett. **75**, 600 (1995), arXiv:gr-qc/9412071.
[45] M. D. Duez, P. Marronetti, S. L. Shapiro, and T. W. Baumgarte, Hydrodynamic simulations in 3+1 general relativity, Phys. Rev. D **67**, 024004 (2003), arXiv:gr-qc/0209102.
[46] Z. B. Etienne, Improved moving-puncture techniques for compact binary simulations, Phys. Rev. D **110**, 064045 (2024), arXiv:2404.01137 [gr-qc].
[47] J. M. Stone, K. Tomida, C. J. White, and K. G. Felker, The Athena++ Adaptive Mesh Refinement Framework: Design and Magnetohydrodynamic Solvers, ApJS **249**, 4 (2020), arXiv:2005.06651 [astro-ph.IM].
[48] R. Löhner, An adaptive finite element scheme for transient problems in CFD, CMAME **61**, 323 (1987).
[49] S. Rosswog, F. Torsello, and P. Diener, The Lagrangian Numerical Relativity code SPHINCS_BSSN_v1.0 (2023), arXiv:2306.06226 [gr-qc].
[50] C. Gundlach, Pseudospectral apparent horizon finders: An Efficient new algorithm, Phys. Rev. D **57**, 863 (1998), arXiv:gr-qc/9707050.
[51] M. Hannam, S. Husa, F. Ohme, B. Bruegmann, and N. O'Murchadha, Wormholes and trumpets: The Schwarzschild spacetime for the moving-puncture generation, Phys. Rev. D **78**, 064020 (2008), arXiv:0804.0628 [gr-qc].
[52] L. I. Petrich, S. L. Shapiro, and S. A. Teukolsky, Oppenheimer-Snyder collapse with maximal time slicing and isotropic coordinates, Phys. Rev. D **31**, 2459 (1985).
[53] A. N. Staley, T. W. Baumgarte, J. D. Brown, B. Farris, and S. L. Shapiro, Oppenheimer-Snyder Collapse in Moving-Puncture Coordinates, Class. Quant. Grav. **29**, 015003 (2012), arXiv:1109.0546 [gr-qc].
[54] S. L. Shapiro and S. A. Teukolsky, Formation of naked singularities: The violation of cosmic censorship, Phys. Rev. Lett. **66**, 994 (1991).
[55] R. M. Wald and V. Iyer, Trapped surfaces in the Schwarzschild geometry and cosmic censorship, Phys. Rev. D **44**, R3719 (1991).
[56] C.-M. Yoo, T. Harada, and H. Okawa, 3D Simulation of Spindle Gravitational Collapse of a Collisionless Particle System, Class. Quant. Grav. **34**, 105010 (2017), arXiv:1611.07906 [gr-qc].
[57] S. Chandrasekhar, The Dynamical Instability of Gaseous Masses Approaching the Schwarzschild Limit in General Relativity, Astrophys. J. **140**, 417 (1964), [Erratum: Astrophys.J. 140, 1342 (1964)].
[58] J. L. Friedman, J. R. Ipser, and R. D. Sorkin, Turning point method for axisymmetric stability of rotating relativistic stars, Astrophys. J. **325**, 722 (1988).